

Module-3

Memory

1

- Basic memory circuits
- Organization of the main memory
- Memory technology
- Cache memory, which reduces the effective memory access time
- Virtual memory, which increases the apparent size of the main memory
- Magnetic and optical disks used for secondary storage

□ **Text Books:**

- Carl Hamacher, Zvonko Vranesic, Safwat Zaky: Computer Organization, 5th Edition, Tata McGraw Hill, 2002.
- Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian : Computer Organization and Embedded Systems, 6th Edition, Tata McGraw Hill, 2012.

□ **Reference Books:**

- William Stallings: Computer Organization & Architecture, 9th Edition, Pearson, 2015.

Introduction

3

- Execution speed of programs is highly dependent on the speed with which instructions and data can be transferred between the processor and the memory.
- Ideally, the memory would be fast, large, and inexpensive.
- Unfortunately, it is impossible to meet all three of these requirements simultaneously. Increased speed and size are achieved at increased cost.
- Much work has gone into developing structures that improve the effective speed and size of the memory, yet keep the cost reasonable.

Basic Concepts-1

4

- The maximum size of the memory that can be used in any computer is determined by the **addressing scheme**.
- For example, a computer that generates 16-bit addresses is capable of addressing up to 2^{16}
= 64K (kilo) memory locations

Basic Concepts-2

5

- The memory is usually designed to **store and retrieve data** in **word-length** quantities.
- Consider, for example, a byte-addressable computer whose instructions generate 32-bit addresses.
- When a 32-bit address is sent from the processor to the memory unit,
 - ✓ the high order 30 bits determine which word will be accessed.
 - ✓ If a byte quantity is specified, the **low-order 2 bits of the address specify which byte location is involved.**

Basic Concepts-3

6

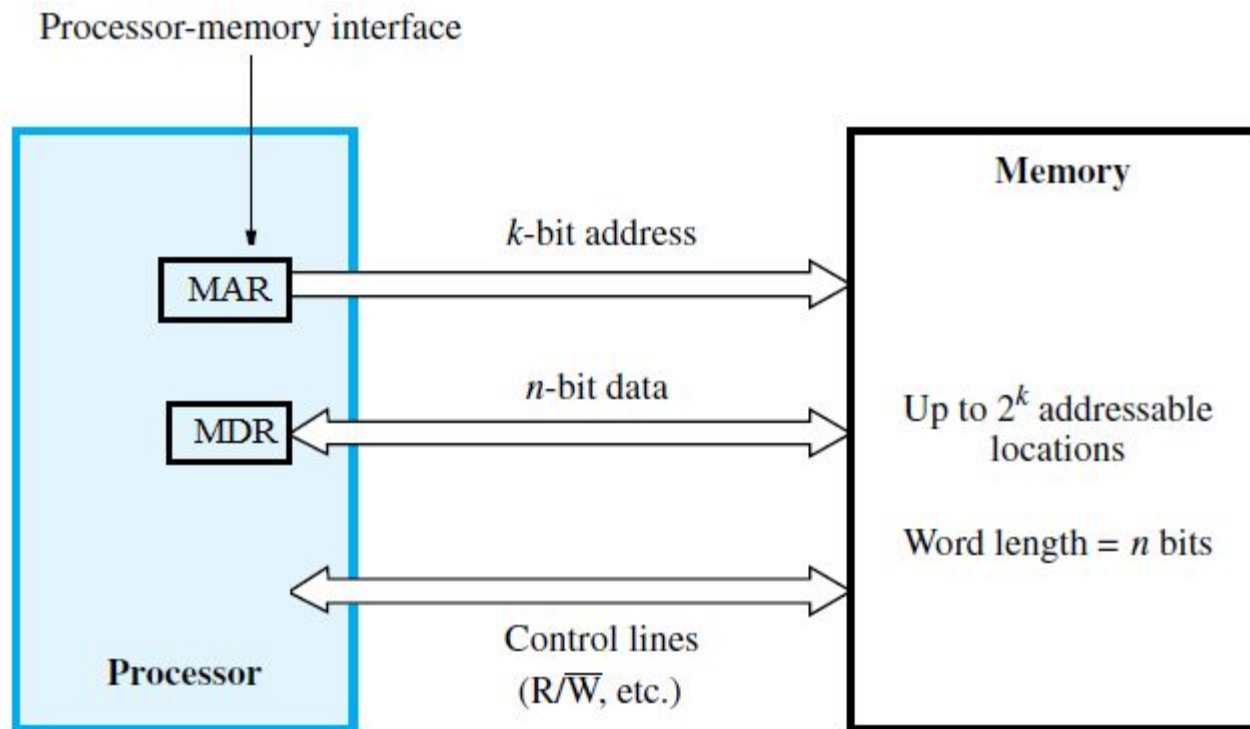
Memory Address (30Bit)	Byte0 (00)	Byte1 (01)	Byte2 (10)	Byte3 (11)
1000 0000H				
1000 0004H				
1000 0008H				
1000 000CH				

- The format (i.e. 2K x 4 or 1M x 1) tells you something about the memory chips' organization.
- The first number (usually the larger number) represents the number of individually addressable units (addresses) and the second number represents the number of bits that are read or written in parallel per addressable unit.
- For example, a 2K x 4 chip (if it exists) would have $2K = 2048 = 2^{11}$ addresses and can write or read 4 bits at a time. This also means this chip would have 11 address lines (maybe multiplexed for a smaller number of external pins) and 4 data lines (the bits that are written or read). To be used in a system that processes 8 bits at a time (like an early home computer), you would need 2 chips in parallel, each providing 4 of the 8 bits.
- A 1M x 1 chip would address 2^{20} individually addressable bits and would usually have a multiplexed address bus to reduce the number of external pins needed on the chip. On a computer with a 32-bit data bus, you would need at least 32 of these working in parallel, each providing 1

Basic Concepts-4

8

- The connection between the processor and its memory consists of address, data, and control lines,



Connection of the memory to the processor.

Basic Concepts-5

- The processor uses the address lines to specify the memory location involved in a data transfer operation, and uses the data lines to transfer the data.
- At the same time, the control lines carry the command indicating a Read or a Write operation and whether a byte or a word is to be transferred. The control lines also provide the necessary timing information and are used by the memory to indicate when it has completed the requested operation.
- When the processor-memory interface receives the memory's response, it asserts the MFC (Memory Function Complete) signal

Basic Concepts-6

10

- A useful measure of the speed of memory units is the **time that elapses** between the **initiation of an operation** to transfer a word of data and the **completion of that operation**.
- This is referred to as the **memory access time**.
- Another important measure is the **memory cycle time**, which is the **minimum time delay required between the initiation of two successive memory operations**, for example, the time between two successive Read operations.
- The cycle time is usually slightly longer than the access time.

Basic Concepts-7

11

- A memory unit is called a *random-access memory* (**RAM**) if the access time to any location is the same, independent of the location's address.
- This distinguishes such memory units from serial, or partly serial, access storage devices such as magnetic and optical disks.
- Access time of the latter devices depends on the address or position of the data.

Basic Concepts-8

12

- **Cache and Virtual Memory**
- The processor of a computer can usually process instructions and data faster than they can be fetched from the main memory. Hence, the memory access time is the bottleneck in the system.
- One way to reduce the memory access time is to use a **cache memory**.
- This is a small, fast memory inserted between the larger, slower main memory and the processor. It holds the currently active portions of a program and their data.
- **Virtual memory** With this technique, only the active portions of a program are stored in the main memory, and the remainder is stored on the much larger secondary storage device. Sections of the program are transferred back and forth between the main memory and the secondary storage device in a manner that is transparent to the application program. As a result, the application program sees a memory that is much larger than the computer's physical main memory.

Semiconductor RAM Memories-1

13

Internal Organization of Memory Chips

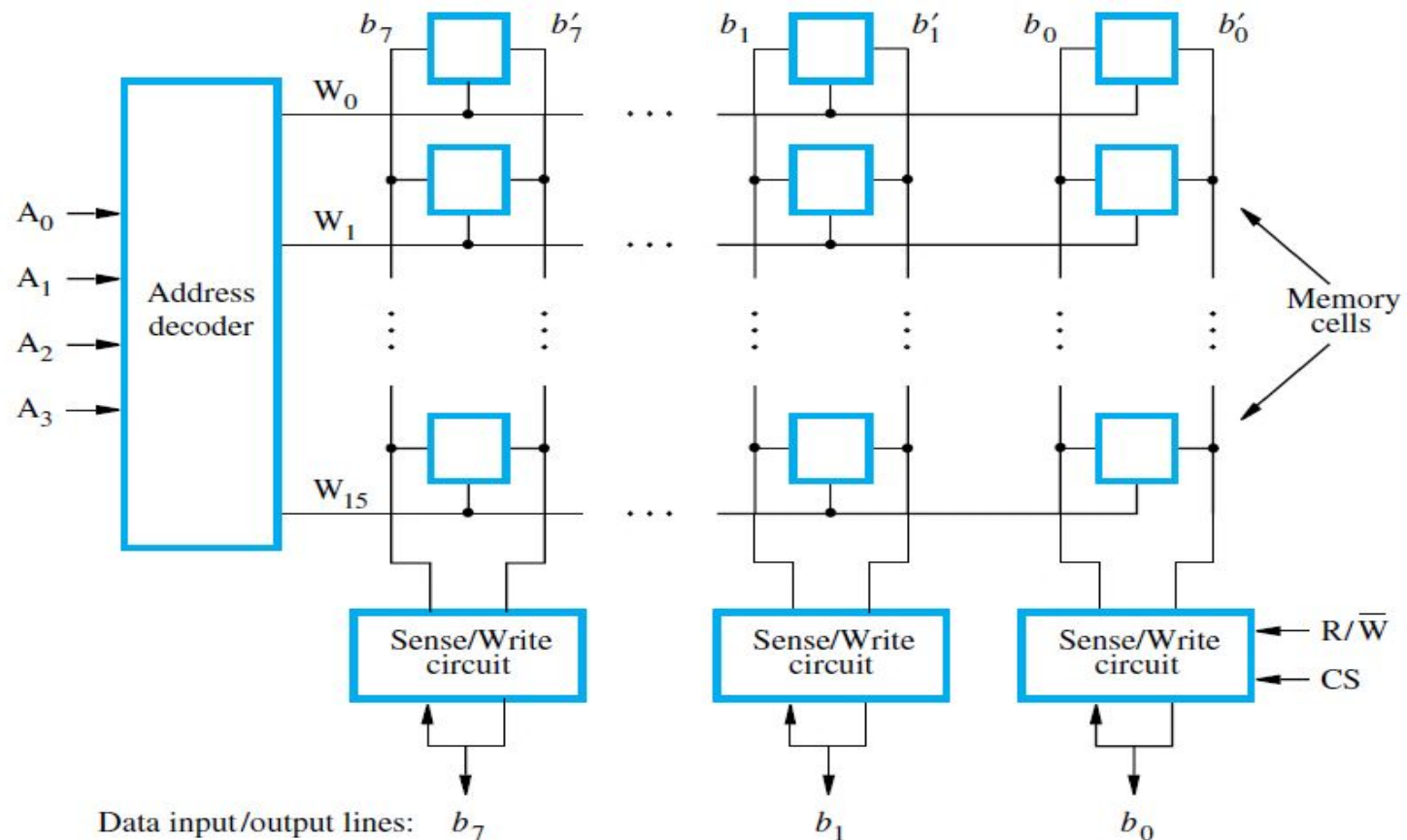


Figure 8.2 Organization of bit cells in a memory chip.

Semiconductor RAM Memories-2

14

- Memory cells are in the form of **an array**, in which each cell is capable of **storing one bit** of information.
- Each row of cells constitutes a memory word, and all cells of a row are connected to a common line referred to as the **word line**, which is driven by the address decoder on the chip.
- The cells in each column are connected to a **Sense/Write circuit** by two *bit lines*, and the Sense/Write circuits are connected to the data input/output lines of the chip.
- During a Read operation, these circuits sense, or read, the information stored in the cells selected by a word line and place this information on the output data lines.
- During a Write operation, the Sense/Write circuits receive input data and store them in the cells of the selected word.

Semiconductor RAM Memories-3

15

- In above example of a very small memory circuit consisting of 16 words of 8 bits each.
- This is referred to as a 16×8 organization.
- The data input and the data output of each Sense/Write circuit are connected to a single bidirectional data line that can be connected to the data lines of a computer.
- The R/W (Read/Write) input specifies the required operation, and the **CS (Chip Select)** input selects a given chip in a multichip memory system.

Semiconductor RAM Memories-4

16

- Number of external connections required
- The memory in above example stores 128 bits and requires 14 external connections for address (4), data(8), and control lines(2). It also needs two lines (2) for power supply and ground connections.
- Now in case of 1K (1024) memory cells. This circuit can be organized as a 128×8 memory, requiring a total of 19 external connections. $(7+8+2+2)$
- The same number of cells can be organized into a $1K \times 1$ format. In this case, a 10-bit address is needed, but there is only one data line, resulting in 15 external connections. $(10+1+2+2)$
- A 1G-bit chip may have a $256M \times 4$ organization, in which case a 28-bit address is needed and 4 bits are transferred to or from the chip. $(28+4+2+2)$

Semiconductor RAM Memories-5

17

- Organization of a $1K \times 1$ memory chip.
- 10-bit address is divided into two groups of 5 bits each to form the row and column addresses for the cell array.
- A row address selects a row of 32 cells, all of which are accessed in parallel.
- But, only one of these cells is connected to the external data line, based on the column address

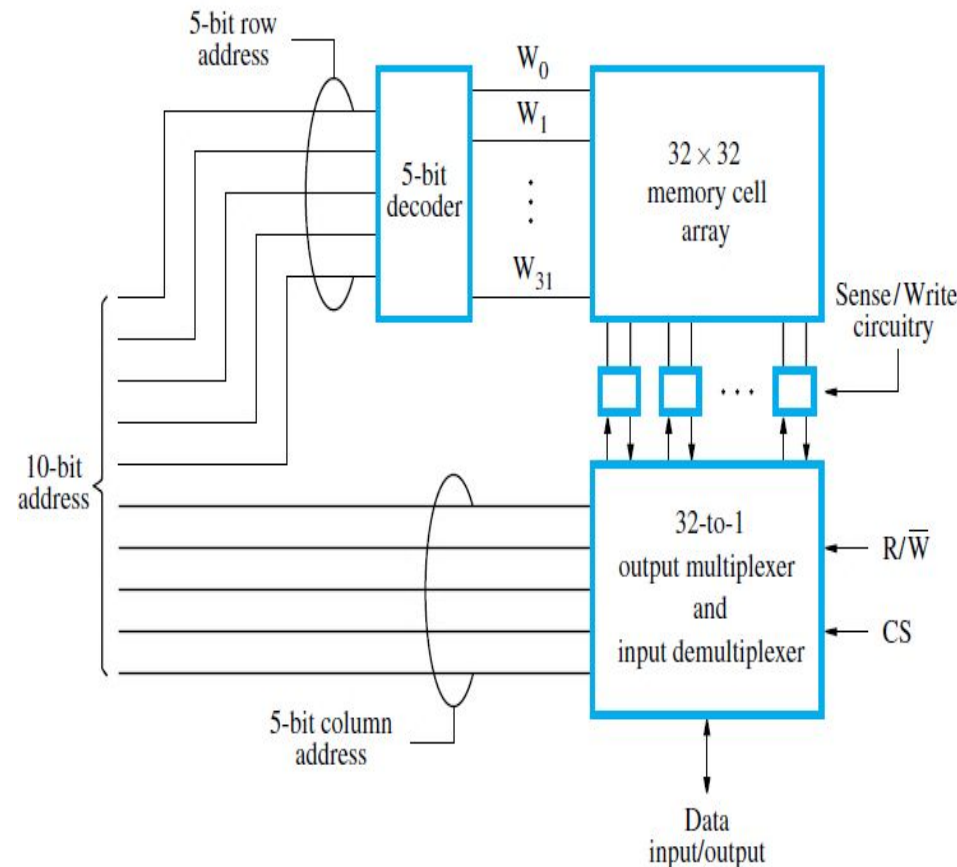


Figure 8.3 Organization of a $1K \times 1$ memory chip.

Semiconductor RAM Memories-6

Static Memories (SRAM)

18

- Memories that consist of circuits **capable of retaining their state as long as power** is applied are known as static memories.

RAM cell operates in 3 modes

- Read
- Write
- Hold

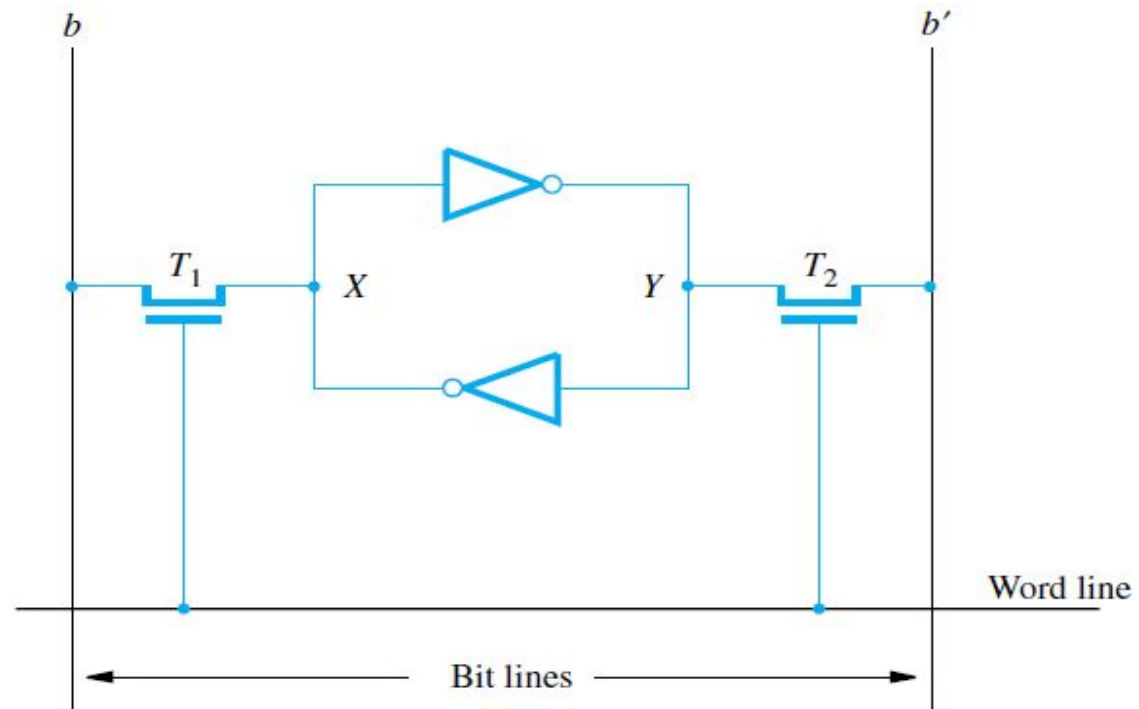


Figure 8.4 A static RAM cell.

Semiconductor RAM Memories-7

Static Memories (SRAM)

19

- Two inverters are cross-connected to form a latch. The latch is connected to two bit lines by transistors $T1$ and $T2$. These transistors act as switches that can be opened or closed under control of the word line.
- When the word line is at ground level, the transistors are turned off and the latch retains its state.
- For example, if the logic value at point X is 1 and at point Y is 0, this state is maintained as long as the signal on the word line is at ground level. Assume that this state represents the value 1.

Access transistors T_1 and T_2 are connected to bit line b and b' , so that we can read from the memory or write into the memory.

for any operation to be performed, the word line should be high

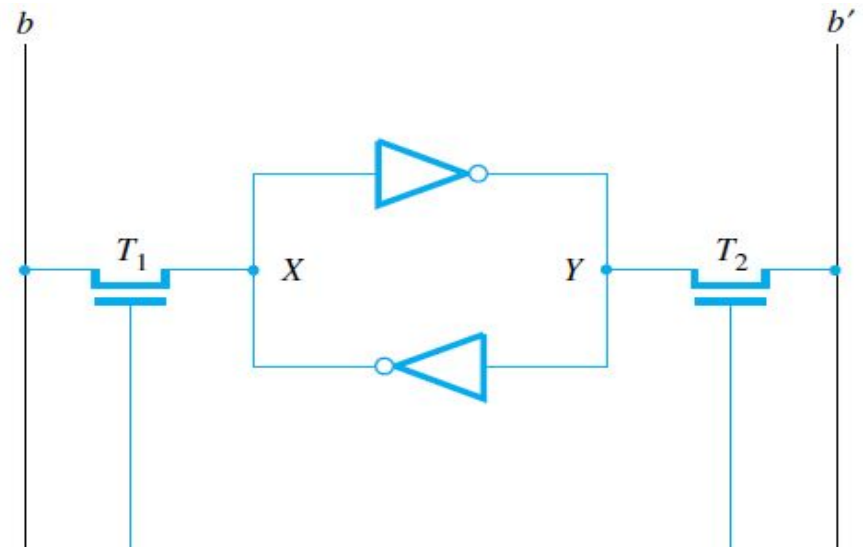
20

Read/Write Mode : If word line is equal to 1, we can access the transistors and hence read and write operations can be performed.

To write into the memory b and b' acts as input, to read from the memory b and b' acts as output lines.

Hold mode : If word line is equal to 0, the access to the transistors will be off and memory will be in hold state.

So as long as SRAM in this mode, the data will remain unchanged .



Read Operation

21

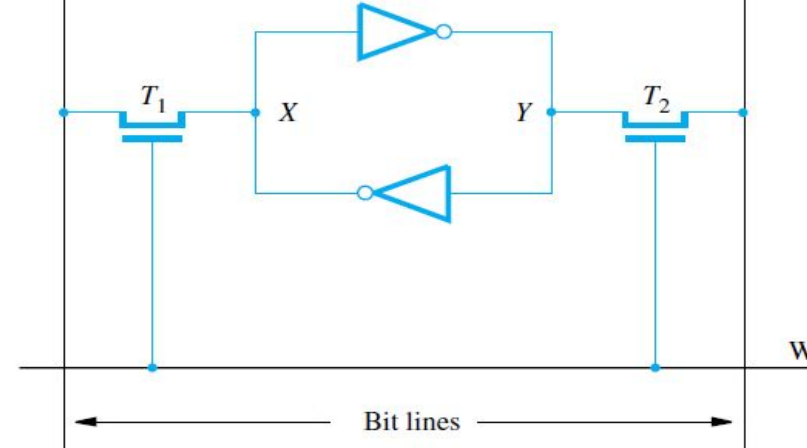


Figure 8.4 A static RAM cell.

1. To perform read operation, initially memory should have some value
2. consider memory has $X=1$ and $Y=0$
3. Raise the word line to high, b and b' acts as output lines that are initially pre-charged (node voltage V_{dd})
4. As X and b are high, there will be no discharge in the circuit.
5. As Y is 0, there will be a voltage difference between the Y and the node voltage at b' , hence b' voltage decreases.
6. Therefore there will be discharge in the circuit and current flows
7. b and b' are connected to the sense amplifier, this sense amplifier acts as a comparator, so When b' is low the output will be 1.
8. Hence input $X=1$ and we got the output as 1, read operation verified
9. In the same way consider $X=0$ and $Y=1$ in the memory.
10. There will be a discharge in the circuit at X and b , since there is voltage difference
11. As bit voltage decreases the output will be 0. when input $X=0$, the output we get is 0.
12. Therefore in both the cases read operation is verified.

Write Operation

22

- Consider the memory bits consists of $X=0$ and $Y=1$. Initially word line is high and hence write operation can be performed.
- In the write operation b and b' are input lines.
- As we have control on the bit lines, initially make the b connected to ground so that we can have the voltage difference between Y and b .
- X will be 1. Initially $X=0$ after the operation $X=1$, hence we write successfully into the memory.

- **Write Operation**
- During a Write operation, the Sense/Write circuit drives bit lines b and b' .
- It places the appropriate value on bit line b and its complement on b' and activates the word line.
- This forces the cell into the corresponding state, which the cell retains when the word line is deactivated.

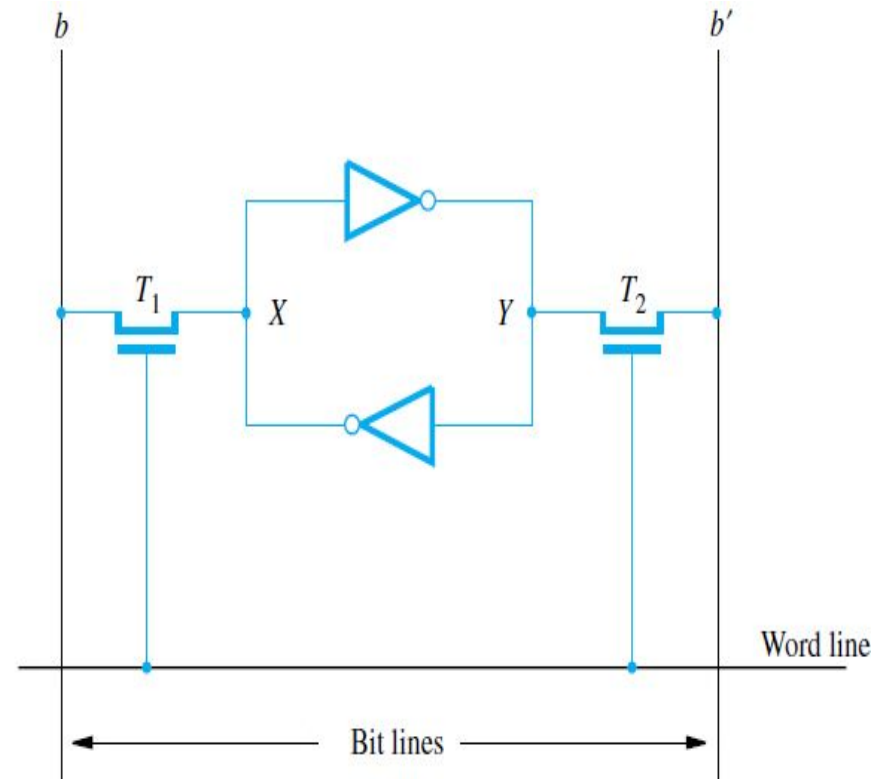


Figure 8.4 A static RAM cell.

Semiconductor RAM Memories-10

Static Memories (SRAM)

24

CMOS Cell

- Transistor pairs (T_3 , T_5) and (T_4 , T_6) form the inverters in the latch.
- The state of the cell is read or written as just explained. For example, in state 1, the voltage at point X is maintained high by having transistors T_3 and T_6 on, while T_4 and T_5 are off.
- If T_1 and T_2 are turned on, bit lines b and b' will have high and low signals, respectively.

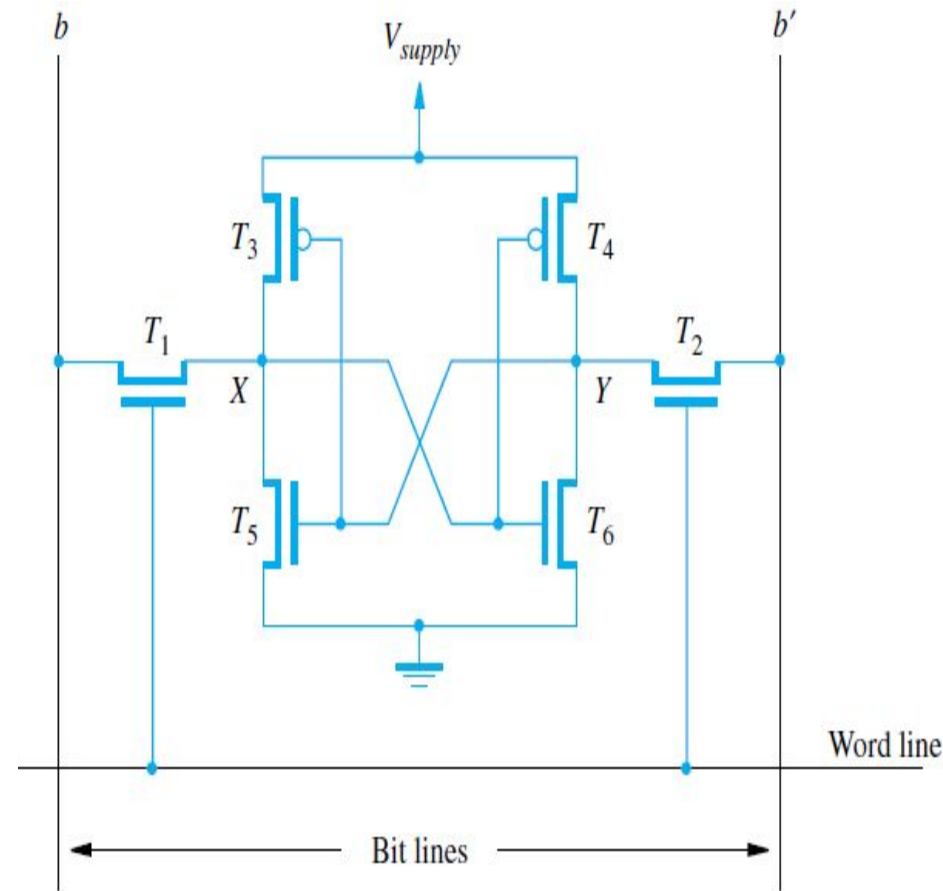


Figure 8.5 An example of a CMOS memory cell.

Semiconductor RAM Memories-1 1

Static Memories (SRAM)

25

- Continuous power is needed for the cell to retain its state. If power is interrupted, the cell's contents are lost. When power is restored, the latch settles into a stable state, but not necessarily the same state the cell was in before the interruption. Hence, SRAMs are said to be **volatile memories**.
- A major advantage of CMOS SRAMs is **their very low power consumption**, because **current flows in the cell only when the cell is being accessed**.
- Static RAMs can be accessed very quickly. Access times on the order of a few nanoseconds are found in commercially available chips.
- SRAMs are used in applications where speed is of critical concern.

Semiconductor RAM Memories-12

Asynchronous Memories (DRAM)

26

- Static RAMs are fast, but their cells require **several transistors**.
- Less expensive and higher density RAMs can be implemented with simpler cells.
- But, these simpler cells do not retain their state for a long period, unless they are accessed frequently (Refreshed) for Read or Write operations.
- Memories that use such cells are called **dynamic RAMs (DRAMs)**.



Semiconductor RAM Memories-13

Asynchronous Memories (DRAM)

27

- Information is stored in a dynamic memory cell in the form of a charge on a capacitor, but this charge can be maintained for only tens of milliseconds.
- Since the cell is required to store information for a much longer time, its contents must be periodically *refreshed (read or write)* by restoring the capacitor charge to its full value.

Semiconductor RAM Memories-14

Asynchronous Memories (DRAM)

28

- A single-transistor dynamic memory cell.
- To store information in this cell, transistor T is turned on and an appropriate voltage is applied to the bit line.
- This causes a known amount of charge to be stored in the capacitor.
- After the transistor is turned off, the charge remains stored in the capacitor, but not for long. The capacitor begins to discharge.

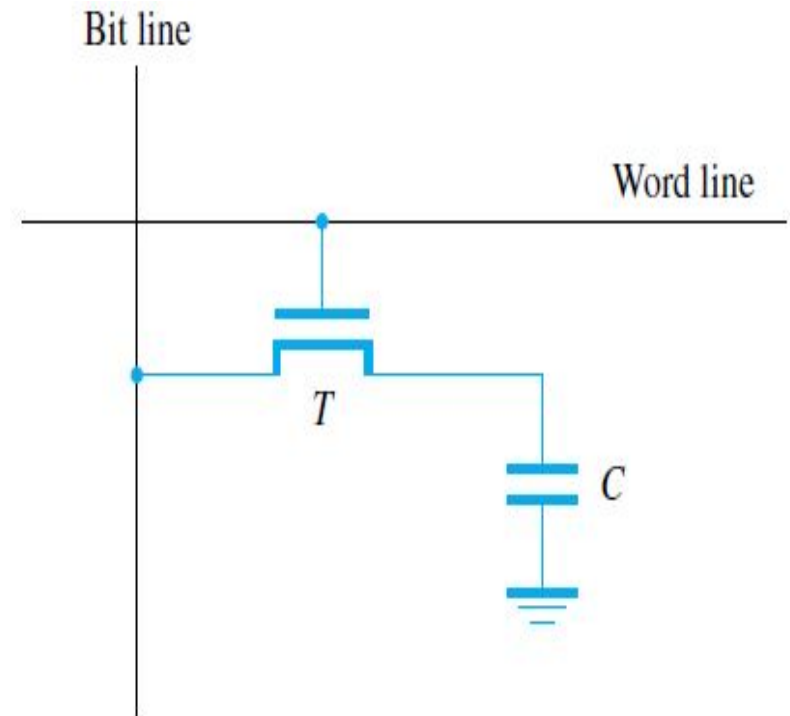


Figure 8.6 A single-transistor dynamic memory cell.

Semiconductor RAM Memories-15

Asynchronous Memories (DRAM)

29

- This is because the transistor continues to conduct a tiny amount of current, after it is turned off.
- Hence, the information stored in the cell can be retrieved correctly only if it is read before the charge in the capacitor drops below some threshold value.
- During a Read operation, the transistor in a selected cell is turned on.
- A sense amplifier connected to the bit line detects whether the charge stored in the capacitor is above or below the threshold value.
- If the charge is above the threshold, the sense amplifier drives the bit line to the full voltage representing the logic value 1. As a result, the capacitor is recharged.
- If the sense amplifier detects that the charge in the capacitor is below the threshold value, it pulls the bit line to ground level to discharge the capacitor fully.
- Thus, reading the contents of a cell automatically refreshes its contents.
- Since the word line is common to all cells in a row, all cells in a selected row are read and refreshed at the same time.

Semiconductor RAM Memories-16

Asynchronous Memories (DRAM)

30

- Internal organization of a $2M \times 8$ dynamic memory chip.

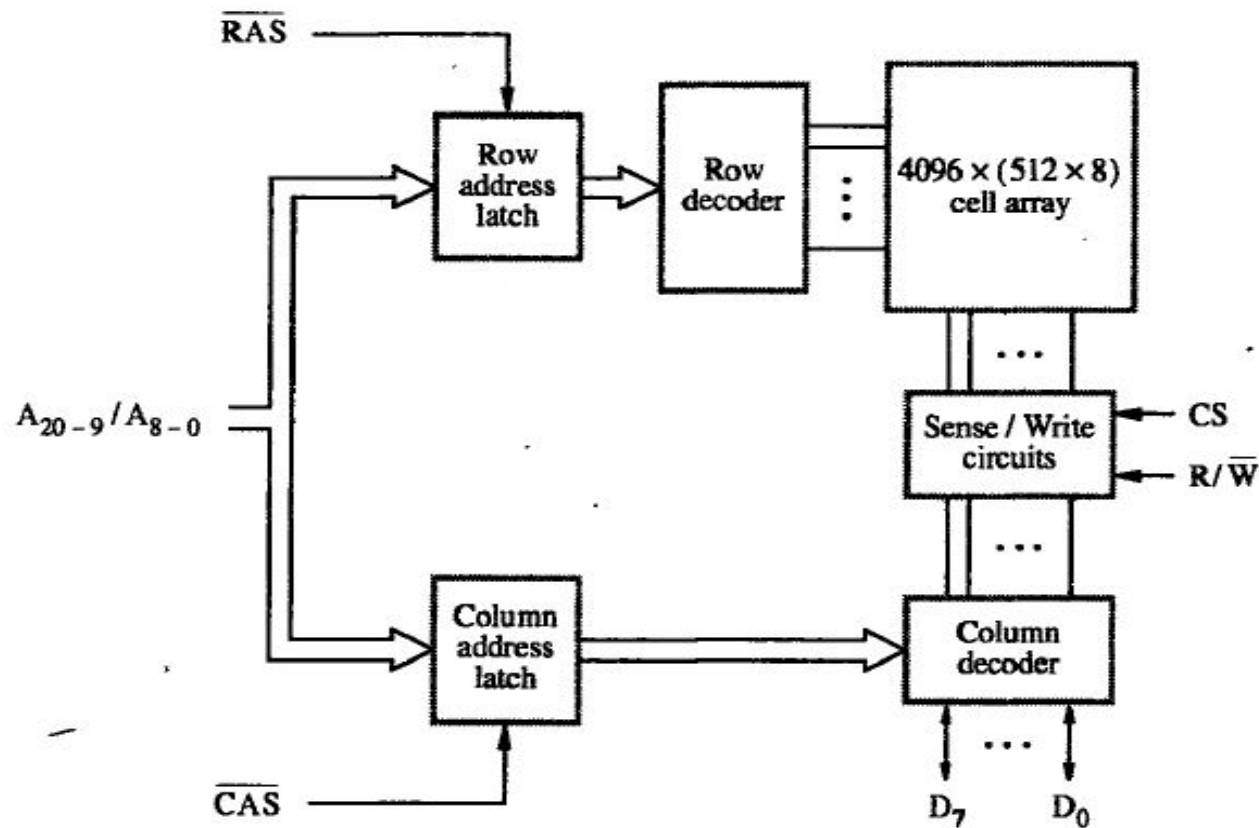


Figure 5.7 Internal organization of a $2M \times 8$ dynamic memory chip.

Semiconductor RAM Memories-17

Asynchronous Memories (DRAM)

31

- A 16-Megabit DRAM chip, configured as $2M \times 8$, is shown in above diagram.
- The cells are organized in the form of a $4K \times 4K$ array.
- The 4096 cells in each row are divided into 512 groups of 8, forming 512 bytes of data.
- Therefore, 12 address bits are needed to select a row, and another 9 bits are needed to specify a group of 8 bits in the selected row.
- In total, a 21-bit address is needed to access a byte in this memory.
- The high-order 12 bits and the low-order 9 bits of the address constitute the row and column addresses of a byte, respectively.
- To reduce the number of pins needed for external connections, the row and column addresses are multiplexed on 12 pins.

Semiconductor RAM Memories-18

Asynchronous Memories (DRAM)

32

- During a Read or a Write operation, the row address is applied first.
- It is loaded into the row address latch in response to a signal pulse on an input control line called the **Row Address Strobe (RAS)**.
- This causes a Read operation to be initiated, in which all cells in the selected row are read and refreshed.

Semiconductor RAM Memories-19

Asynchronous Memories (DRAM)

33

- Shortly after the row address is loaded, the **column address is applied** to the address pins and loaded into the column address latch under control of a second control line called the **Column Address Strobe (CAS)**.
- The information in this latch is decoded and the appropriate group of 8 Sense/Write circuits is selected.
- If the R/W control signal indicates a Read operation, the output values of the selected circuits are transferred to the data lines, D7–D0.
- For a Write operation, the information on the D7–D0 lines is transferred to the selected circuits, then used to overwrite the contents of the selected cells in the corresponding 8 columns.
- We should note that in commercial DRAM chips, the RAS and CAS control signals are active when low.

Semiconductor RAM Memories-20

Asynchronous Memories (DRAM)

34

- The timing of the operation of the DRAM described above is controlled by the RAS and CAS signals.
- These signals are generated by a **memory controller circuit external** to the chip when the processor issues a Read or a Write command.
- During a Read operation, the output data are transferred to the processor after a delay equivalent to the memory's access time.
- Such memories are referred to as *asynchronous DRAMs*.
- The memory controller is also responsible for refreshing the data stored in the memory chips.

Semiconductor RAM Memories-21

Asynchronous Memories (DRAM)

35

- **Fast Page Mode**
- When the DRAM in above example is accessed, the contents of all 4096 cells in the selected row are sensed, but only 8 bits are placed on the data lines, D7–D0. This byte is selected by the column address, bits A8–0.
- A simple addition to the circuit makes it possible to access the other bytes in the same row without having to reselect the row.
- Each sense amplifier also acts as a latch.
- When a row address is applied, the contents of all cells in the selected row are loaded into the corresponding latches.
- Then, it is only necessary to apply different column addresses to place the different bytes on the data lines.

Semiconductor RAM Memories-22

Asynchronous Memories (DRAM)

36

- This arrangement leads to a very useful feature. **All bytes in the selected row can be transferred in sequential order by applying a consecutive sequence of column addresses under the control of successive CAS signals.**
- Thus, a block of data can be transferred at a much faster rate than can be achieved for transfers involving random addresses.
- This feature is referred to as the ***fast page mode***. (A large block of data is often called a page.)

Semiconductor RAM Memories-23

Synchronous DRAMs (SDRAM)

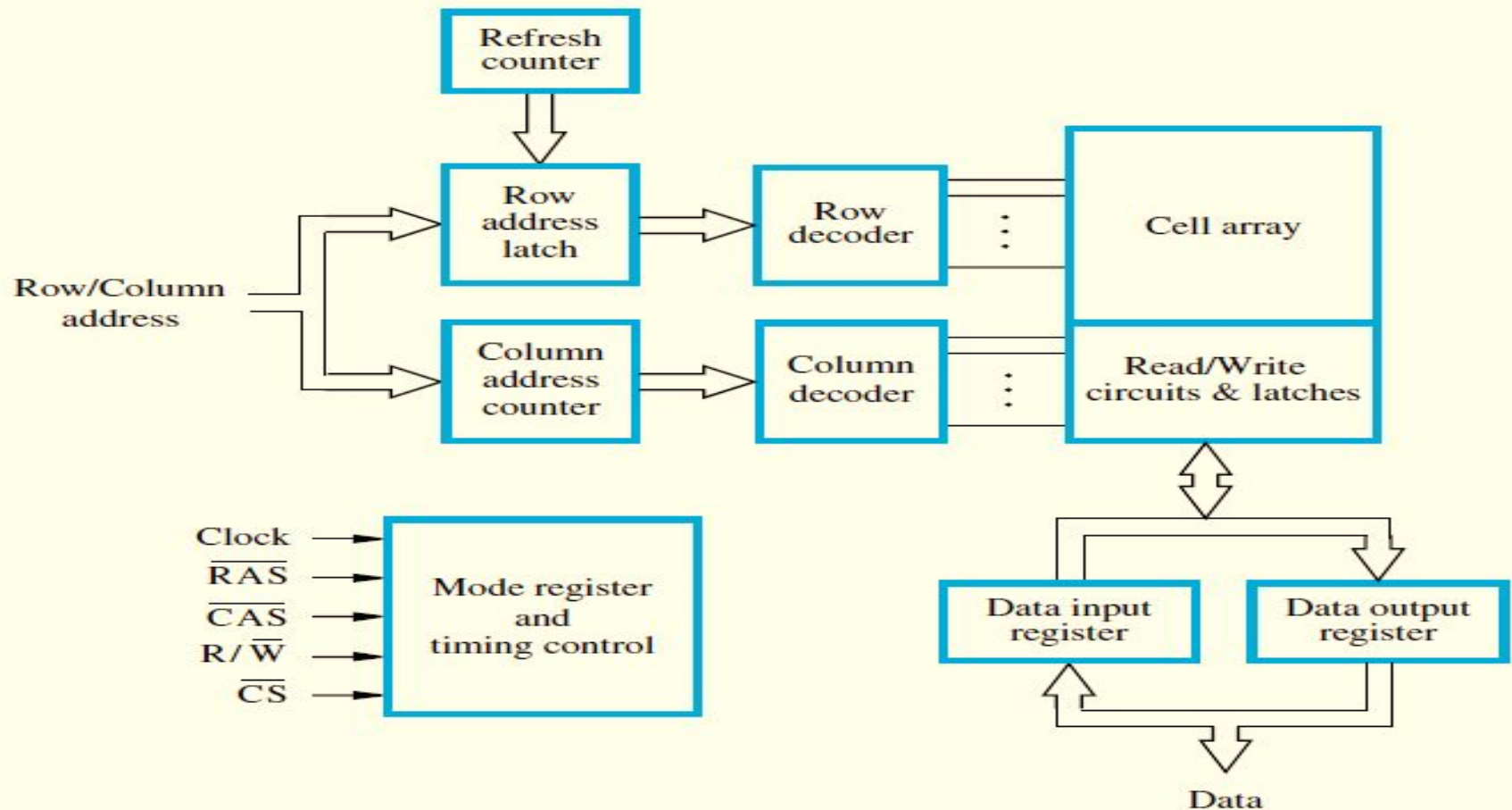
37

- **DRAMs whose operation is synchronized with a clock signal.** Such memories are known as *synchronous DRAMs* (SDRAMs).
- The cell array is the same as in asynchronous DRAMs.
- **SDRAMs have built-in refresh circuitry, with a refresh counter to provide the addresses of the rows to be selected for refreshing.**
- As a result, the dynamic nature of these memory chips is almost invisible to the user.

Semiconductor RAM Memories-24

Synchronous DRAMs(SDRAM)

38



Synchronous DRAM.

Semiconductor RAM Memories-25

Synchronous DRAMs(SDRAM)

39

- SDRAMs have several different modes of operation, which can be selected by writing **control information into a *mode register***.
- For example, in **burst operations** of different lengths can be specified.
- It is not necessary to provide externally-generated pulses on the CAS line to select successive columns.
- The necessary control signals are generated internally using a **column counter and the clock signal**.
- New data are placed on the data lines **at the rising edge of each clock pulse**.

Semiconductor RAM Memories-26

Synchronous DRAMs(SDRAM)

40

- Timing diagram for a typical burst read of length 4

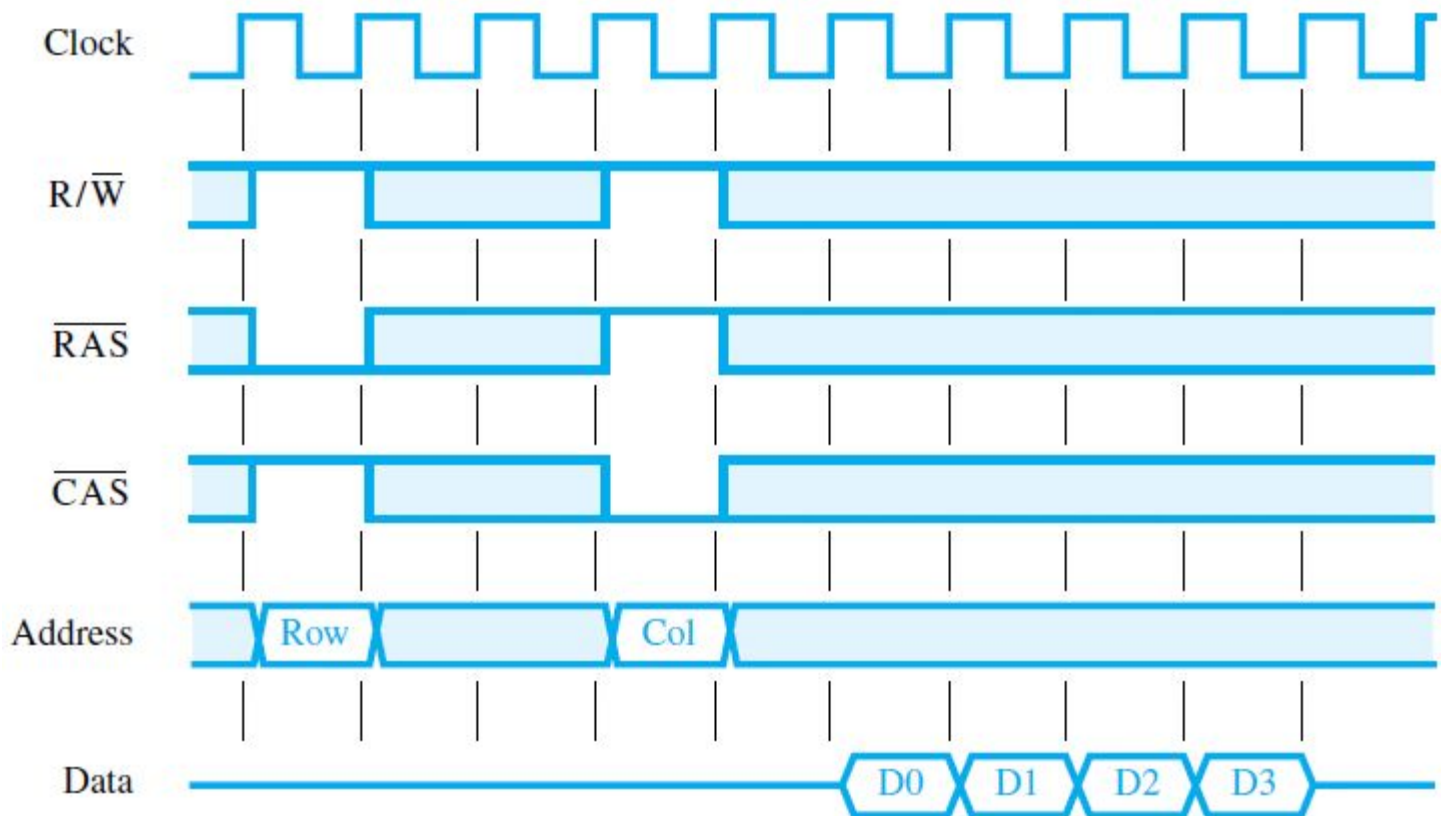


Figure 8.9 A burst read of length 4 in an SDRAM.

Semiconductor RAM Memories-27

Synchronous DRAMs(SDRAM)

41

- First, the row address is latched under control of the RAS signal. The memory typically takes 2 or 3 clock cycles (we use 2 in the figure for simplicity) to activate the selected row. Then, the column address is latched under control of the CAS signal.
- After a delay of one clock cycle, the first set of data bits is placed on the data lines.
- The SDRAM automatically increments the column address to access the next three sets of bits in the selected row, which are placed on the data lines in the next 3 clock cycles.

Semiconductor RAM Memories-28

Synchronous DRAMs(SDRAM)

42

- Synchronous DRAM can deliver data at a very high rate, because all the control signals needed are generated inside the chip.
- The initial commercial SDRAMs designed for clock speeds of up to 133 MHz.
- As technology evolved, much faster SDRAM chips were developed. Today's SDRAMs operate with clock speeds that can exceed 1 GHz.

Semiconductor RAM Memories-28

Synchronous DRAMs(SDRAM)

43

- **Latency and Bandwidth**
- Data transfers to and from the main memory often involve blocks of data.
- **The memory access time defined earlier is not sufficient for describing the memory's performance when transferring blocks of data.** During block transfers, *memory latency* is the amount of time it takes to transfer the first word of a block.
- The time required to transfer a complete block depends also on the rate at which successive words can be transferred and on the size of the block. **The time between successive words of a block is much shorter than the time needed to transfer the first word.**
- For instance, as in the timing diagram, the access cycle begins with the assertion of the RAS signal. The first word of data is transferred five clock cycles later. Thus, the latency is five clock cycles. If the clock rate is 500 MHz, then the latency is 10 ns.
- The remaining three words are transferred in consecutive clock cycles, at the rate of one word every 2 ns.

Semiconductor RAM Memories-29

Synchronous DRAMs(SDRAM)

44

- **A useful performance measure is the number of bits or bytes that can be transferred in one second. This measure is referred as the memory *bandwidth*.**
- It depends on the speed of access to the stored data and on the number of bits that can be accessed in parallel.
- The rate at which data can be transferred to or from the memory depends on the bandwidth of the system interconnections.
- For this reason, the interconnections used always ensure that the **bandwidth available for data transfers between the processor and the memory is very high.**

Semiconductor RAM Memories-30

Synchronous DRAMs

45

- ❑ **Double-Data-Rate SDRAM (DDR-RAM)**
- ❑ The key idea is to take advantage of the fact that a large number of bits are accessed at the same time inside the chip when a row address is applied.
- ❑ **To make the best use of the available clock speed, data are transferred externally on both the rising and falling edges of the clock.**
- ❑ For this reason, memories that use this technique are called *double-data-rate SDRAMs* (DDR SDRAMs).

Semiconductor RAM Memories-31

Synchronous DRAMs

46

- Several versions of DDR chips have been developed. The earliest version is known as DDR. Later versions, called DDR2, DDR3, and DDR4, have enhanced capabilities.
- They offer increased storage capacity, lower power, and faster clock speeds.
- For example, DDR2 and DDR3 can operate at clock frequencies of 400 and 800 MHz, respectively. Therefore, they transfer data using the effective clock speeds of 800 and 1600 MHz, respectively.

Semiconductor RAM Memories-32

Structure of Larger Memories

47

- The basic organization of memory circuits as they may be implemented on **a single chip**.
- Now examine how memory chips may be connected to form a much larger memory.

Semiconductor RAM Memories-33

Structure of Larger Memories

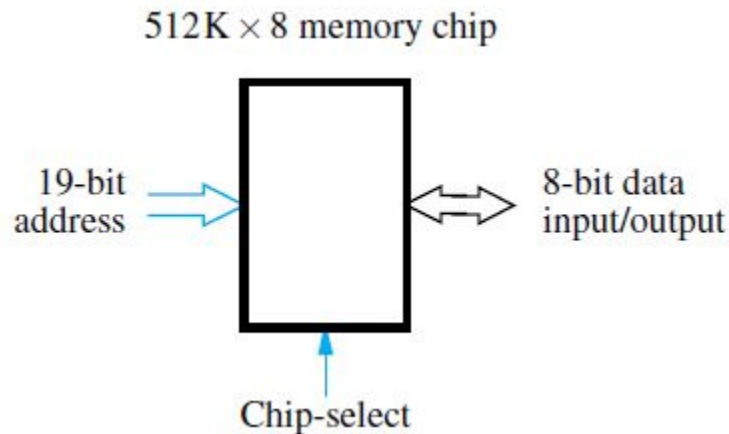
48

- ❑ **Static Memory Systems**
- ❑ Consider a memory consisting of $2M$ words of 32 bits each. Shown in next slide, how this memory can be implemented using $512K \times 8$ static memory chips.
- ❑ Each column in the figure implements one byte position in a word, with four chips providing $2M$ bytes.
- ❑ Four columns implement the required $2M \times 32$ memory.

Semiconductor RAM Memories-34

Structure of Larger Memories

49



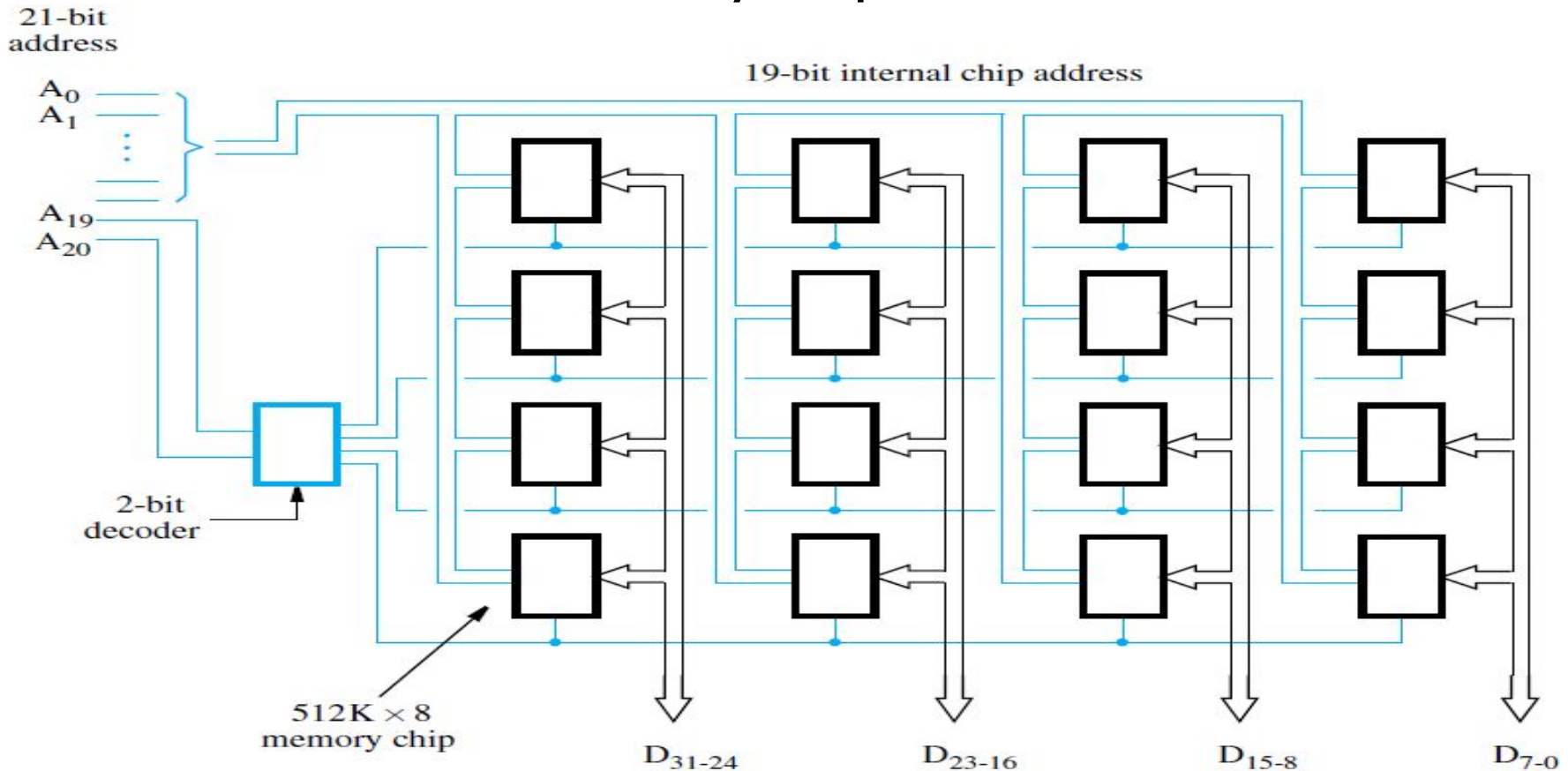
- Each chip has a control input called Chip-select. When this input is set to 1, it enables the chip to accept data from or to place data on its data lines.

Semiconductor RAM Memories-35

Structure of Larger Memories

50

- Organization of a $2M \times 32$ memory module using $512K \times 8$ static memory chips.



Semiconductor RAM Memories-36

Structure of Larger Memories

51

- Only the selected chip places data on the data output line, while all other outputs are electrically disconnected from the data lines.
- 21 address bits are needed to select a 32-bit word in this memory.
- The high-order two bits of the address are decoded to determine which of the four rows should be selected.
- The remaining 19 address bits are used to access specific byte locations inside each chip in the selected row.
- The R/W inputs of all chips are tied together to provide a common Read/Write control line (not shown in the figure).

Semiconductor RAM Memories-37

Structure of Larger Memories

52

- ❑ **Dynamic Memory Systems**
- ❑ Modern computers use very large memories. Even a small personal computer is likely to have at least 1G bytes of memory. Typical desktop computers may have 4G bytes or more of memory.
- ❑ A large memory leads to better performance, because more of the programs and data used in processing can be held in the memory, **thus reducing the frequency of access to secondary storage.**

Semiconductor RAM Memories-38

Structure of Larger Memories

53

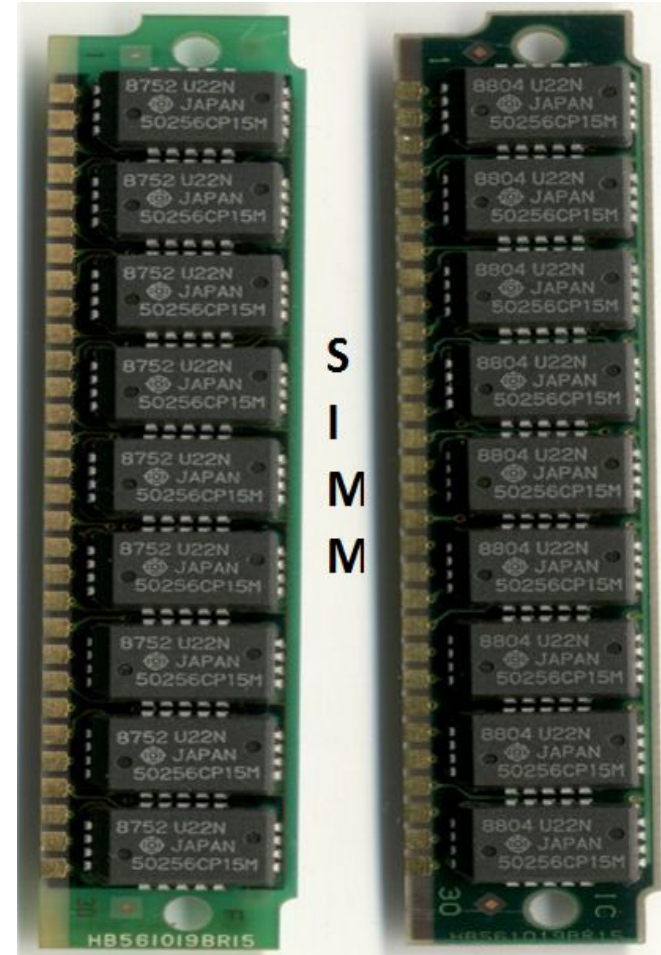
- Because of their high bit density and low cost, dynamic RAMs, mostly of the synchronous type, are widely used in the memory units of computers.
- They are slower than static RAMs, but they use less power and have considerably lower cost per bit.
- Available chips have capacities as high as 2G bits, and even larger chips are being developed. To reduce the number of memory chips needed in a given computer, a memory chip may be organized to read or write a number of bits in parallel.
- Chips are manufactured in different organizations, to provide flexibility in designing memory systems. For example, a 1-Gbit chip may be organized as $256M \times 4$, or $128M \times 8$.

Semiconductor RAM Memories-39

Structure of Larger Memories

54

- **Packaging considerations have led to the development of assemblies known as memory modules.** Each such module houses many memory chips, typically in the range 16 to 32, on a small board that plugs into a socket on the computer's motherboard.
- **Memory modules are commonly called *SIMMs* (Single In-line Memory Modules) or *DIMMs* (Dual In-line Memory Modules), depending on the configuration of the pins.**
- Modules of different sizes are designed to use the same socket.
- For example, $128\text{M} \times 64$, $256\text{M} \times 64$, and $512\text{M} \times 64$ bit DIMMs all use the same 240-pin socket.
- Thus, total memory capacity is easily expanded by replacing a smaller module with a larger one, using the same socket.

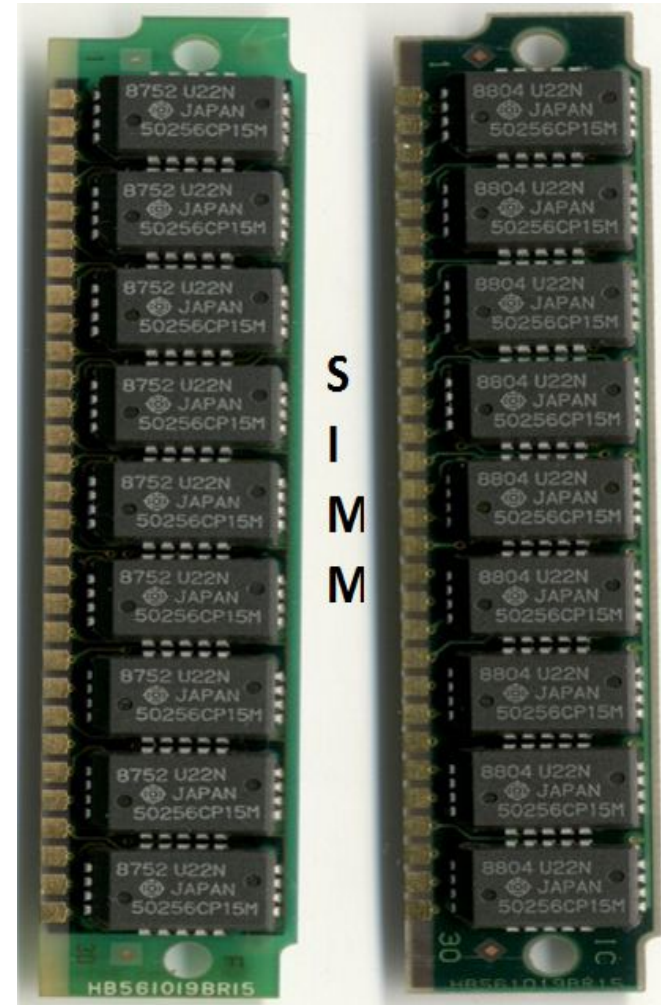


Semiconductor RAM Memories-39

Structure of Larger Memories

55

- **Packaging considerations have led to the development of assemblies known as memory modules.** Each such module houses many memory chips, typically in the range 16 to 32, on a small board that plugs into a socket on the computer's motherboard.
- **Memory modules are commonly called *SIMMs* (Single In-line Memory Modules) or *DIMMs* (Dual In-line Memory Modules), depending on the configuration of the pins.**
-
- Modules of different sizes are designed to use the same socket.
- For example, $128\text{M} \times 64$, $256\text{M} \times 64$, and $512\text{M} \times 64$ bit DIMMs all use the same 240-pin socket.
- Thus, total memory capacity is easily expanded by replacing a smaller module with a larger one, using the same socket.



Semiconductor RAM Memories-40

Structure of Larger Memories

56

□ **Memory Controller**

- The address applied to dynamic RAM chips is divided into two parts , i.e. the high-order address bits, which select a row in the cell array by activating the RAS signal. Then, the low-order address bits, which select a column, are provided on the same address pins and latched under control of the CAS signal.
- Since a typical processor issues all bits of an address at the same time, a multiplexer is required. This function is usually performed by a *memory controller* circuit.
- The controller accepts a complete address and the R/W signal from the processor, under control of a *Request* signal which indicates that a memory access operation is needed.
- It forwards the R/W signals and the row and column portions of the address to the memory and generates the RAS and CAS signals, with the appropriate timing.

Semiconductor RAM Memories-41

Structure of Larger Memories

57

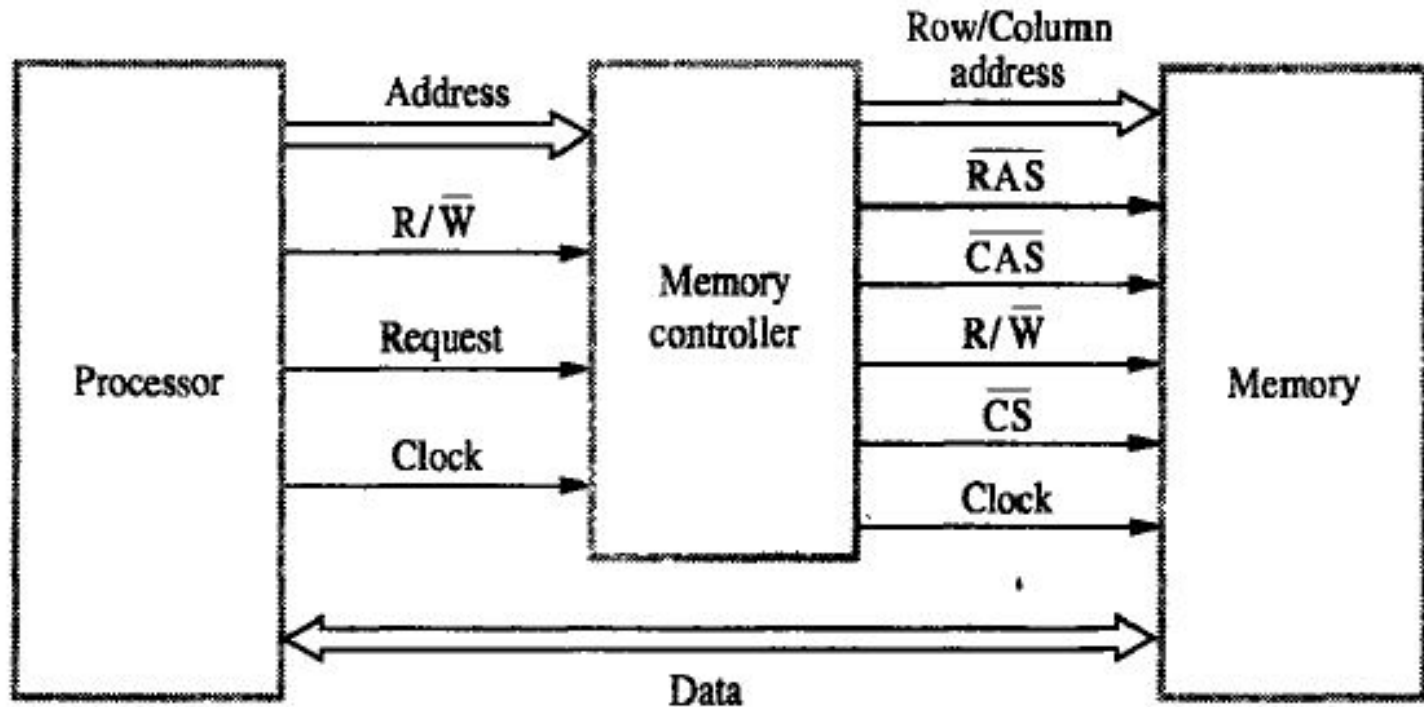


Figure 5.11 Use of a memory controller.

Semiconductor RAM Memories-42

Structure of Larger Memories

58

- **Memory Controller**
- When a memory includes multiple modules, one of these modules is selected based on the high-order bits of the address.
- The memory controller decodes these high-order bits and generates the chip-select signal for the appropriate module.
- Data lines are connected directly between the processor and the memory.

Semiconductor RAM Memories-43

Structure of Larger Memories

59

- **Memory Controller**
- Dynamic RAMs must be refreshed periodically. The circuitry required to initiate refresh cycles is included as part of the internal control circuitry of synchronous DRAMs.
- However, a control circuit external to the chip is needed to initiate periodic Read cycles to refresh the cells of an asynchronous DRAM.
- The memory controller provides this capability.

Read-only Memories-1

60

- Both static and dynamic RAM chips are volatile, which means that they retain information only while power is turned on.
- There are many applications requiring memory devices that retain the stored information when power is turned off. For example, need to store a small program in such a memory, to be used to start the bootstrap process of loading the operating system from a hard disk into the main memory. The embedded applications are another important example.
- Many embedded applications do not use a hard disk and require non-volatile memories to store their software.
- Different types of non-volatile memories have been developed. But, a special writing process is needed to place the information into a non-volatile memory. Since its normal operation involves only reading the stored data, a memory of this type is called a ***read-only memory (ROM)***.

Read-only Memories-2

ROM

61

- A memory is called a read-only memory, or ROM, when information can be written into it only once at the time of manufacture.

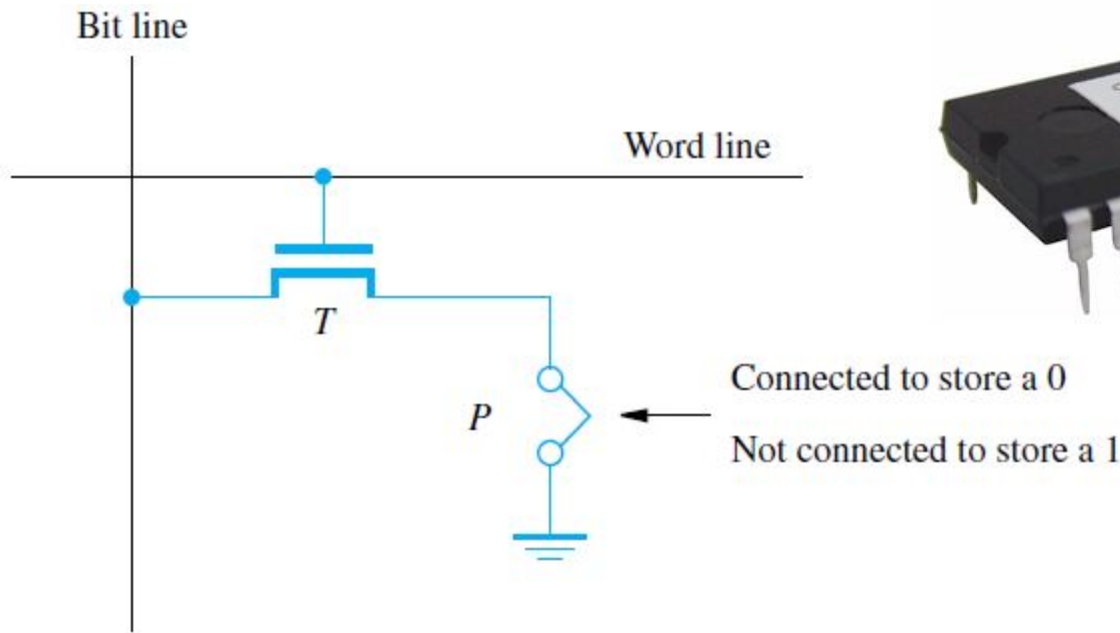


Figure 8.11 A ROM cell.

Read-only Memorie

ROM

62

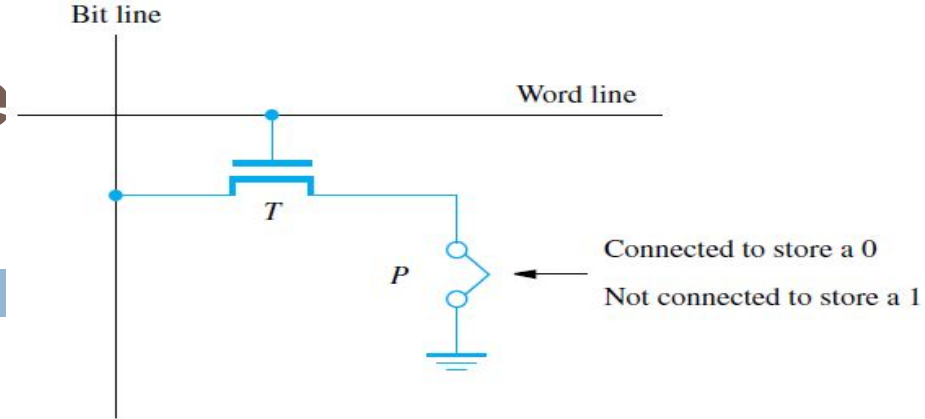
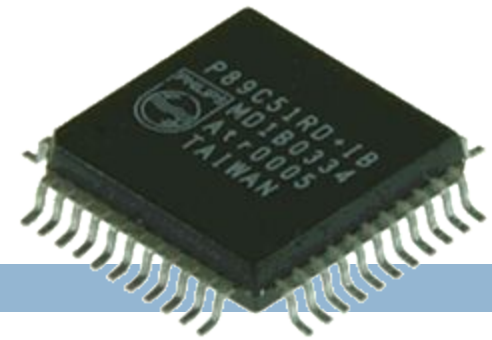


Figure 8.11 A ROM cell.

- A logic value 0 is stored in the cell if the transistor is connected to ground at point *P*; otherwise, a 1 is stored.
- The bit line is connected through a resistor to the power supply. To read the state of the cell, the word line is activated to close the transistor switch.
- As a result, the voltage on the bit line drops to near zero if there is a connection between the transistor and ground.
- If there is no connection to ground, the bit line remains at the high voltage level, indicating a 1.
- A sense circuit at the end of the bit line generates the proper output value. The state of the connection to ground in each cell is determined when the chip is manufactured, using a mask with a pattern that represents the information to be stored.

Read-only Memories-4

PROM



63

- Some ROM designs allow the **data to be loaded by the user**, thus providing a programmable ROM (PROM).
- **Programmability is achieved by inserting a fuse at point P in a ROM cell.**
- Before it is programmed, the memory contains all 0s.
- The user can insert 1s at the required locations by burning out the fuses at these locations using high-current pulses. **This process is irreversible.**
- PROMs provide flexibility and convenience not available with ROMs. The cost of preparing the masks needed for storing a particular information pattern makes ROMs cost effective only in large volumes.
- The alternative technology of PROMs provides a more convenient and considerably less expensive approach, because memory chips can be programmed directly by the user.

Read-only Memories-5

EPROM



64

- It allows the stored data to be erased and new data to be written into it. Such an **erasable, reprogrammable ROM** is usually called an *EPROM*.
- Since EPROMs are capable of retaining stored information for a long time, they can be used in place of ROMs or PROMs while software is being developed.
- An EPROM cell has a structure similar to the ROM cell. However, the connection to ground **at point P is made through a special transistor**. The transistor is normally turned off, creating an open switch. It can be turned on by injecting charge into it that becomes trapped inside.
- By injecting the charges into it that become trapped in the transistors that form the memory cells. This can be done by exposing the chip to ultraviolet light, which erases the entire contents of the chip. To make this possible, EPROM chips are mounted in packages that have transparent windows.

Read-only Memories-6

EEPROM

65

- An EPROM must be physically removed from the circuit for reprogramming. Also, the stored information cannot be erased selectively. The entire contents of the chip are erased when exposed to ultraviolet light.
- Another type of erasable PROM can be programmed, **erased, and reprogrammed electrically**. Such a chip is called an *electrically erasable* PROM, or EEPROM.
- **One disadvantage of EEPROMs is that different voltages are needed for erasing, writing, and reading the stored data, which increases circuit complexity.**
- However, this disadvantage is outweighed by the many advantages of EEPROMs. They have replaced EPROMs in practice.

Read-only Memories-7

Flash Memory

66

- An approach similar to EEPROM technology has given rise to *flash memory* devices. A flash cell is based on a single transistor controlled by trapped charge, much like an EEPROM cell.
- The key difference is that, in a **flash device**, it is **only possible to write an entire block of cells**. Prior to writing, the previous contents of the block are erased.
- Flash devices have greater density, which leads to higher capacity and a lower cost per bit. They require a single power supply voltage, and consume less power in their operation.

Read-only Memories-8

Flash Memory

67

- The low power consumption of flash memories makes them attractive for use in portable, battery-powered equipment.
- Typical applications include hand-held computers, cell phones, digital cameras, and MP3 music players. A flash memory is used in digital cameras to store picture data. In MP3 players, flash memories store the data that represent sound.
- Single flash chips may not provide sufficient storage capacity for the applications mentioned above. Larger memory modules consisting of a number of chips are used where needed.
- There are two popular choices for the implementation of such modules: **flash cards and flash drives.**

Read-only Memories-9

Flash Memory

68



- **Flash Cards**
- One way of constructing a larger module is to mount flash chips on a small card. Such flash cards have a standard interface that makes them usable in a variety of products.
- A card is simply plugged into a conveniently accessible slot. Flash cards with a USB interface are widely used and are commonly known as memory keys.
- They come in a variety of memory sizes. Larger cards may hold as much as 32 Gbytes. A minute of music can be stored in about 1 Mbyte of memory, using the MP3 encoding format.
- Hence, a 32-Gbyte flash card can store approximately 500 hours of music.

Read-only Memories-10

Flash Memory

69



- **Flash Drives**
- Larger flash memory modules have been developed to replace hard disk drives, and hence are called flash drives.
- They are designed to fully emulate hard disks, to the point that they can be fitted into standard disk drive bays. However, the storage capacity of flash drives is significantly lower.
- Currently, the capacity of flash drives is on the order of 64 to 128 Gbytes. In contrast, hard disks have capacities exceeding a terabyte. Also, disk drives have a very low cost per bit.
- The fact that flash drives are solid state electronic devices with no moving parts provides important advantages over disk drives. They have shorter access times, which result in a faster response. They are insensitive to vibration and they have lower power consumption, which makes them attractive for portable, battery-driven applications.

Direct Memory Access – Refer text book

Speed, Size and Cost-1

71

- A very fast memory can be implemented using static RAM chips. But, these chips are not suitable for implementing large memories, because their basic cells are larger, consume more power than dynamic RAM cells and not cost effective.

Speed, Size and Cost-2

72

- Although dynamic memory units with gigabyte capacities can be implemented at a reasonable cost, the affordable size is still small compared to the demands of large programs with voluminous data.
- A solution is provided by using secondary storage, mainly magnetic disks, to provide the required memory space. Disks are available at a reasonable cost, and they are used extensively in computer systems. However, they are much slower than semiconductor memory units.
- So, a very large amount of cost-effective storage can be provided by magnetic disks, and a large and considerably faster, yet affordable, main memory can be built with dynamic RAM technology.
- This leaves the more expensive and much faster static RAM technology to be used in smaller units where speed is of the essence, such as in cache memories.

Speed, Size and Cost-3

73

Memory hierarchy

- The fastest access is to data held in processor registers.
- *Processor cache*, holds copies of the instructions and data stored in a much larger memory that is provided externally.
- A primary cache is always located on the processor chip. This cache is small and its access time is comparable to that of processor registers. The primary cache is referred to as the *level 1 (L1)* cache.
- A larger, and hence somewhat slower, secondary cache is placed between the primary cache and the rest of the memory. It is referred to as the *level 2 (L2)* cache.

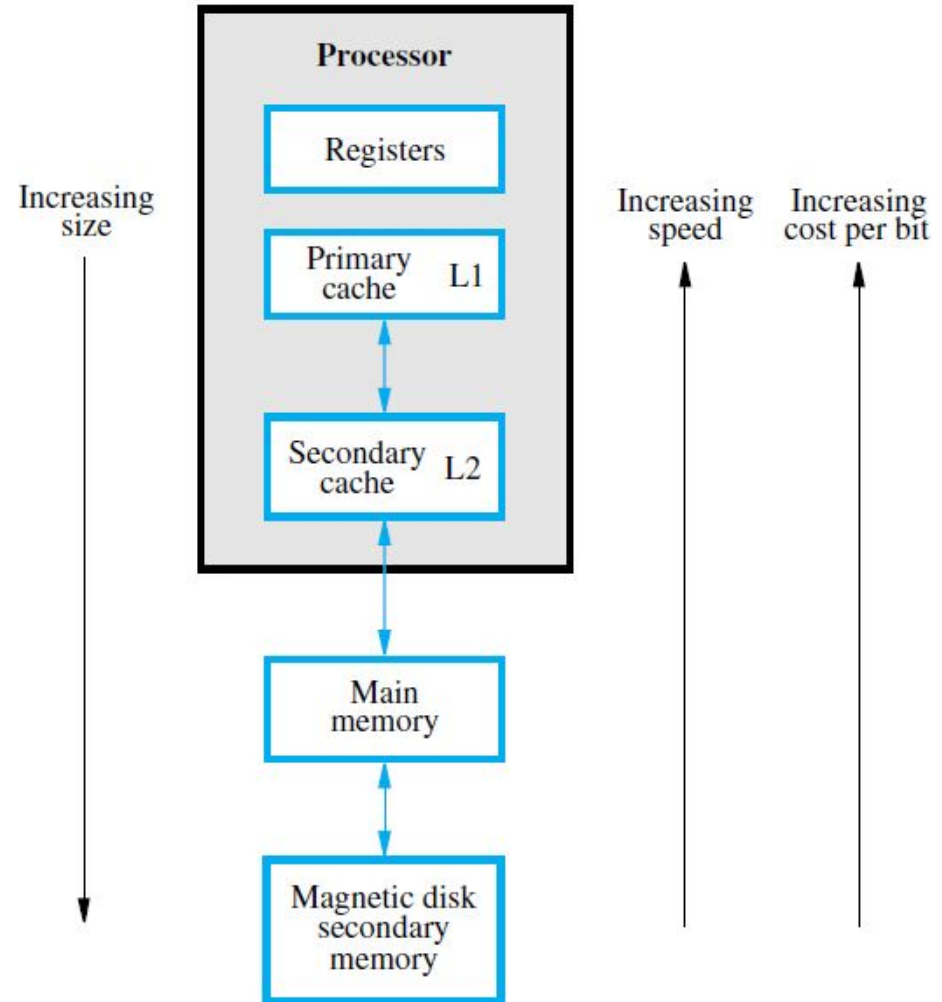


Figure 8.14 Memory hierarchy.

Speed, Size and Cost-4

74

- Some computers have a *level 3* (L3) cache of even larger size, in addition to the L1 and L2 caches. An L3 cache, also implemented in SRAM technology, may or may not be on the same chip with the processor and the L1 and L2 caches.
- The *main memory*. This is a large memory implemented using dynamic memory components, typically assembled in memory modules such as DIMMs or SIMM or RIMM.
- The main memory is much larger but significantly slower than cache memories.
- Disk devices provide a very large amount of inexpensive memory, and they are widely used as secondary storage in computer systems. They are very slow compared to the main memory.
- During program execution, the speed of memory access is of utmost importance.
- **The key to managing the operation of the hierarchical memory system is to bring the instructions and data that are about to be used as close to the processor as possible. This is the main purpose of using cache memories.**

Cache Memories-1

75

- The cache is a small and very fast memory, interposed between the processor and the main memory.
- Its purpose is to make the main memory appear to the processor to be much faster than it actually is.
- The effectiveness of this approach is based on a property of computer programs called ***locality of reference***.

Cache Memories-2

76

- Analysis of programs shows that most of their execution time is spent in routines in which many instructions are executed repeatedly.
- The actual detailed pattern of instruction sequencing is not important—the point is that many instructions in localized areas of the program are executed repeatedly during some time period.
- This behaviour manifests itself in two ways: **temporal and spatial**.
- **Temporal** - recently executed instruction is likely to be executed again very soon.
- **The spatial-** means that instructions close to a recently executed instruction are also likely to be executed soon.

Cache Memories-3

77

- The memory control circuitry is designed to take advantage of the property of locality of reference.
- Temporal locality suggests that whenever an information item, instruction or data, is first needed, this item should be brought into the cache, because it is likely to be needed again soon.
- Spatial locality suggests that instead of fetching just one item from the main memory to the cache, it is useful to fetch several items that are located at adjacent addresses as well.
- The term **cache block or cache line** refers to a set of contiguous address locations of some size.

Cache Memories-4

78

- Use of a cache memory.

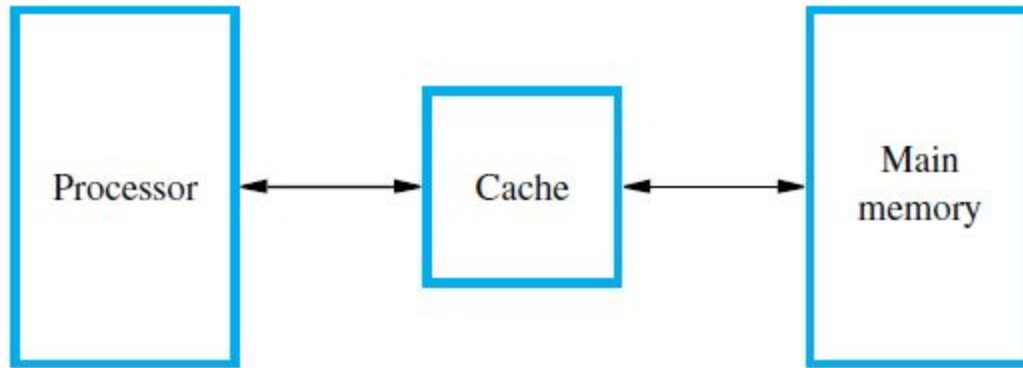


Figure 8.15 Use of a cache memory.

- When the processor issues a Read request, the contents of a block of memory words containing the location specified are transferred into the cache.

Cache Memories-5

79

- Subsequently, when the program references any of the locations in this block, the desired contents are read directly from the cache.
-
- Usually, the cache memory can store a reasonable number of blocks at any given time, but this number is small compared to the total number of blocks in the main memory.
- The correspondence between the main memory blocks and those in the cache is specified by a ***mapping function***.
- When the cache is full and a memory word (instruction or data) that is not in the cache is referenced, the cache control hardware must decide which block should be removed to create space for the new block that contains the referenced word. The collection of rules for making this decision constitutes the cache's ***replacement algorithm***.

Cache Memories-6

80

□ Cache Hits

- The processor does not need to know explicitly about the existence of the cache. It simply issues R/W requests using addresses that refer to locations in the memory.
- The cache control circuitry determines whether the requested word currently exists in the cache. If it does, the R/W operation is performed on the appropriate cache location. In this case, a *read* or *write hit* is said to have occurred. The main memory is not involved when there is a **cache hit** in a Read/Write operation.

Cache Memories-7

81

- For a Write operation, the system can proceed in one of two ways.
- In the **first technique**, called the ***write-through protocol***, both the cache location and the main memory location are updated.
- The **second technique** is to update only the cache location and to **mark the block containing it with an associated flag bit**, often called the ***dirty or modified bit***.
 - The main memory location of the word is updated later, when the block containing this marked word is removed from the cache to make room for a new block. **This technique is known as the *write-back, or copy-back, protocol*.**

Cache Memories-8

82

- The write-through protocol is simpler than the write-back protocol, but it results in unnecessary Write operations in the main memory when a given cache word is updated several times during its cache residency.
- The write-back protocol also involves unnecessary Write operations, because all words of the block are eventually written back, even if only a single word has been changed while the block was in the cache.
- The write-back protocol is used most often, to take advantage of the high speed with which data blocks can be transferred to memory chips.

Cache Memories-9

83

- **Cache Misses**
- A Read operation for a word that is not in the cache constitutes a *Read miss*.
- It causes the block of words containing the requested word to be copied from the main memory into the cache. After the entire block is loaded into the cache, the particular word requested is forwarded to the processor.
- **Alternatively, this word may be sent to the processor as soon as it is read from the main memory.**
- The latter approach, which is called *load-through*, or *early restart*, reduces the processor's waiting time somewhat, at the expense of more complex circuitry.

Cache Memories-10

84

- When a *Write miss* occurs in a computer that uses the write-through protocol, the information is written directly into the main memory.
- For the write-back protocol, the block containing the addressed word is first brought into the cache, and then the desired word in the cache is overwritten with the new information.

Cache Memories-1 1

85

- In a pipelined processor can cause instruction execution to stall for one or more cycles.
- This can occur if a Load or Store instruction requests access to data in the memory at the same time that a subsequent instruction is being fetched. When this happens, instruction fetch is delayed until the data access operation is completed.
- To avoid stalling the pipeline, many processors use separate caches for instructions and data, making it possible for the two operations to proceed in parallel.

Cache Memories-12

Mapping Functions

86

- There are several possible methods for determining where memory blocks are placed in the cache.
- Consider a example, cache consisting of 128 blocks of 16 words each, for a total of 2048 (2K) words, and assume that the main memory is addressable by a 16-bit address.
- The main memory has 64K words, which will view as 4K blocks of 16 words each.
- For simplicity, assume that consecutive addresses refer to consecutive words.

Cache Memories-13

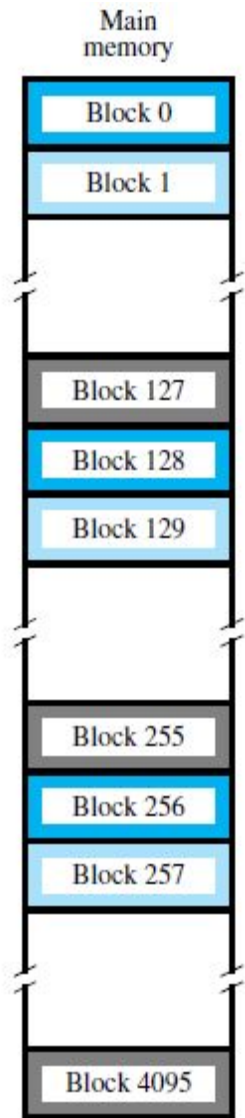
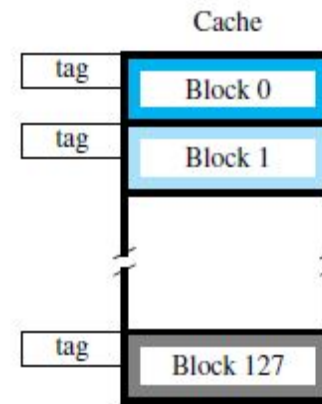
Mapping Functions

87

- **Direct Mapping**

- The simplest way to determine cache locations in which to store memory blocks is the *direct-mapping* technique.
- In this technique, block j of the main memory maps onto block $j \bmod 128$ of the cache, as shown in next slide.
- Thus, whenever one of the main memory blocks 0, 128, 256, ... is loaded into the cache, it is stored in cache block 0. Blocks 1, 129, 257, ... are stored in cache block 1, and so on.
- Since more than one memory block is mapped onto a given cache block position, contention may arise for that position even when the cache is not full.
- For example, instructions of a program may start in block 1 and continue in block 129, possibly after a branch. As this program is executed, both of these blocks must be transferred to the block-1 position in the cache. Contention is resolved by allowing the new block to overwrite the currently resident block.

88



Tag	Block	Word
5	7	4

Main memory address

Figure 8.16 Direct-mapped cache.

Cache Memories-15

Mapping Functions

89

- With direct mapping, the replacement algorithm is trivial. Placement of a block in the cache is determined by its memory address.
- The memory address can be divided into three fields, as shown in previous slide.
- The low-order 4 bits select one of 16 words in a block. When a new block enters the cache, the 7-bit cache block field determines the cache position in which this block must be stored.

Cache Memories-16

Mapping Functions

90

- The high-order 5 bits of the memory address of the block are stored in 5 *tag* bits associated with its location in the cache.
- The tag bits identify which of the 32 main memory blocks mapped into this cache position is currently resident in the cache.
- As execution proceeds, the 7-bit cache block field of each address generated by the processor points to a particular block location in the cache.
- The high-order 5 bits of the address are compared with the tag bits associated with that cache location. If they match, then the desired word is in that block of the cache. If there is no match, then the block containing the required word must first be read from the main memory and loaded into the cache.
- The direct-mapping technique is easy to implement, but it is not very flexible.

Cache Memories-17

Mapping Functions

91

- **Associative Mapping**
- In which a main memory block can be placed into any cache block position.
- In diagram (Next slide), 12 tag bits are required to identify a memory block when it is resident in the cache.
- The tag bits of an address received from the processor are compared to the tag bits of each block of the cache to see if the desired block is present. This is called the *associative-mapping* technique.

Cache Memories-18

Mapping Functions

92

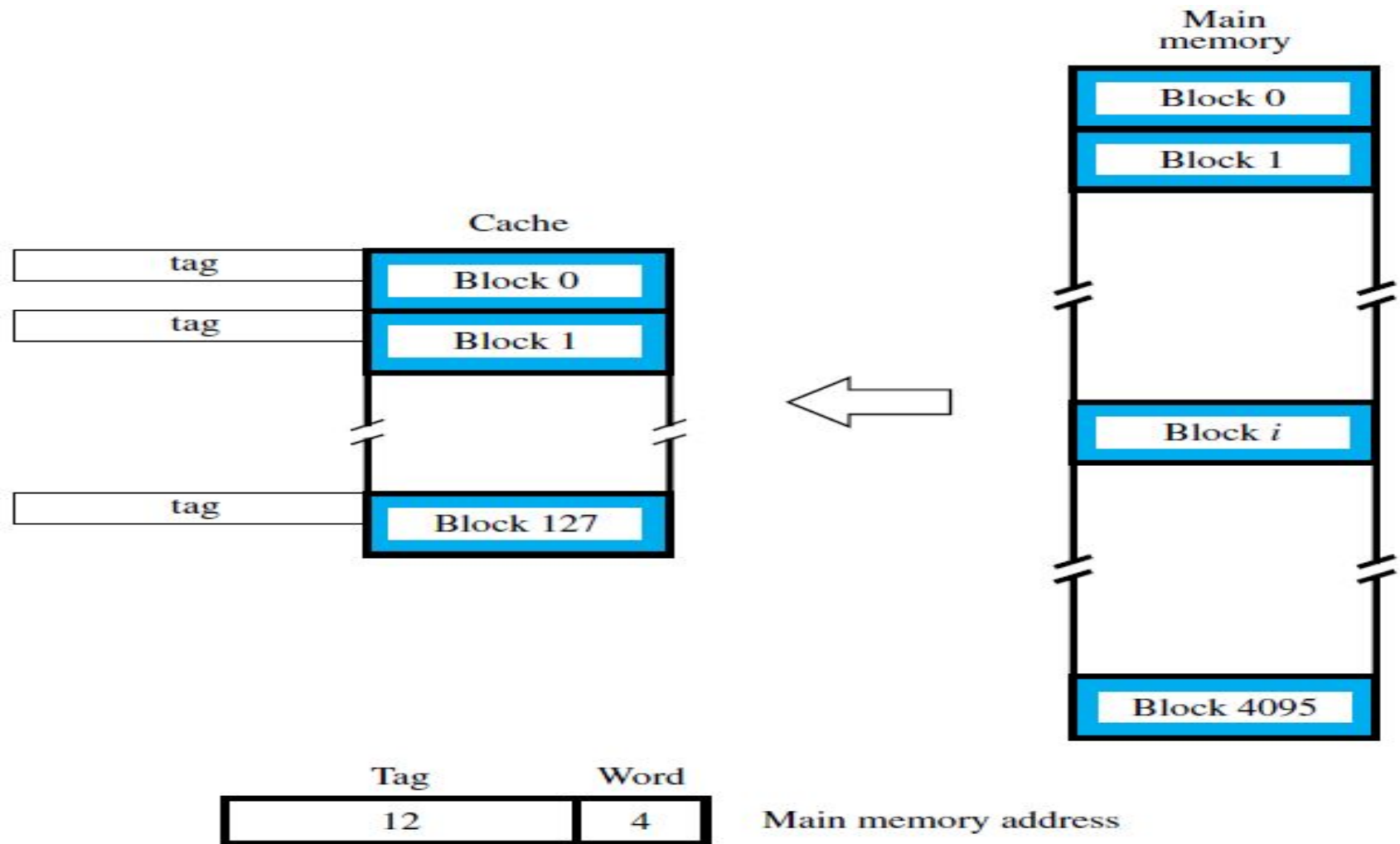


Figure 8.17 Associative-mapped cache.

Cache Memories-19

Mapping Functions

93

- It gives complete freedom in choosing the cache location in which to place the memory block, resulting in a more efficient use of the space in the cache.
- When a new block is brought into the cache, it replaces (ejects) an existing block only if the cache is full.
- In this case, we need an algorithm to select the block to be replaced.
- The complexity of an associative cache is higher than that of a direct-mapped cache, because of the need to search all 128 tag patterns to determine whether a given block is in the cache.
- To avoid a long delay, the tags must be searched in parallel. A search of this kind is called an *associative search*.

Cache Memories-20

Mapping Functions

94

- **Set-Associative Mapping**
- Another approach is to use a combination of the direct- and associative-mapping techniques.
- The blocks of the cache are grouped into sets, and the mapping allows a block of the main memory to reside in any block of a specific set.
- Hence, the contention problem of the direct method is eased by having a few choices for block placement.
- At the same time, the hardware cost is reduced by decreasing the size of the associative search.

Cache Memories-21

Mapping Functions

95

- Memory blocks 0, 64, 128, . . . , 4032 map into cache set 0, and they can occupy either of the two block positions within this set.
- Having 64 sets means that the 6-bit set field of the address determines which set of the cache might contain the desired block.

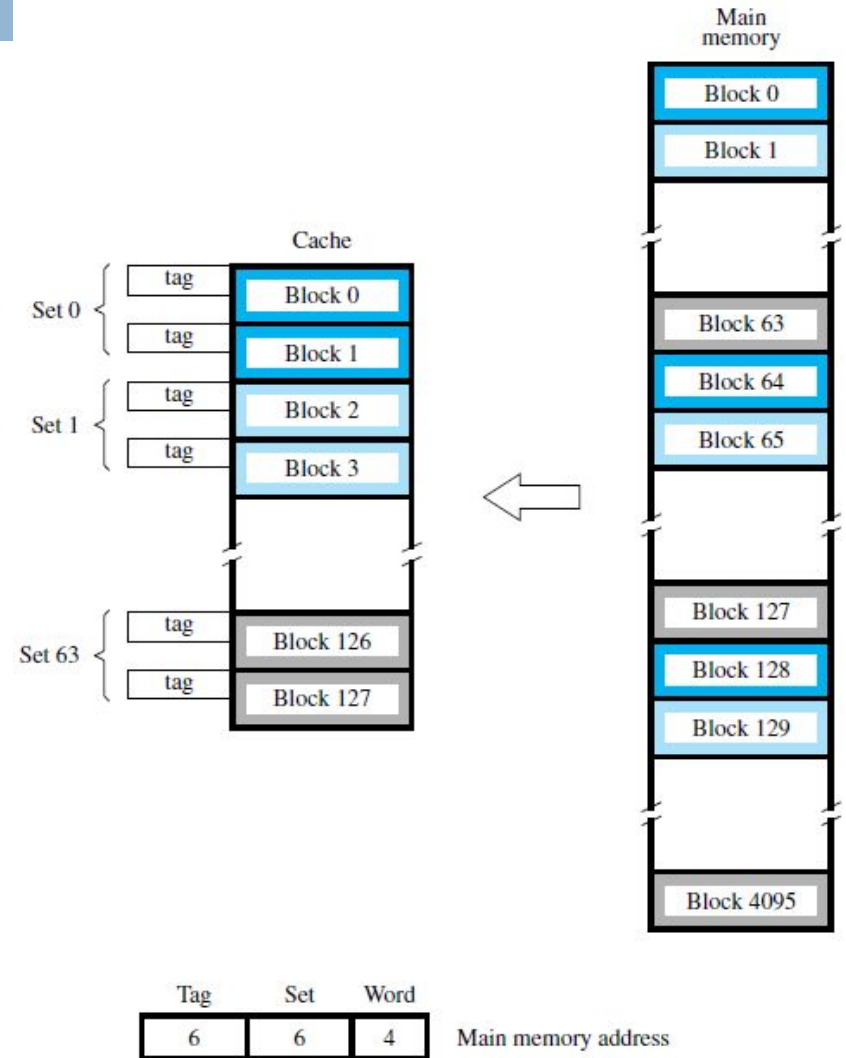


Figure 8.18 Set-associative-mapped cache with two blocks per set.

Cache Memories-22

Mapping Functions

96

- The tag field of the address must then be associatively compared to the tags of the two blocks of the set to check if the desired block is present. This two-way associative search is simple to implement.
- The number of blocks per set is a parameter that can be selected to suit the requirements of a particular computer.
- For the main memory and cache sizes in above example, four blocks per set can be accommodated by a 5-bit set field, eight blocks per set by a 4-bit set field, and so on.
- The extreme condition of 128 blocks per set requires no set bits and corresponds to the fully-associative technique, with 12 tag bits. The other extreme of one block per set is the direct-mapping method.
- A cache that has k blocks per set is referred to as a k -way set-associative cache.

Cache Memories-23

97

□ **Stale Data**

- When power is first turned on, the cache contains no valid data. A control bit, usually called the *valid bit*, must be provided for each cache block to indicate whether the data in that block are valid.
- The valid bits of all cache blocks are set to 0 when power is initially applied to the system. Some valid bits may also be set to 0 when new programs or data are loaded from the disk into the main memory.
- Data transferred from the disk to the main memory using the DMA mechanism are usually loaded directly into the main memory, bypassing the cache.
- If the memory blocks being updated are currently in the cache, the valid bits of the corresponding cache blocks are set to 0.
- As program execution proceeds, the valid bit of a given cache block is set to 1 when a memory block is loaded into that location.
- The processor fetches data from a cache block only if its valid bit is equal to 1. The use of the valid bit in this manner ensures that the processor will not fetch *stale (old)* data from the cache.

Cache Memories-24

98

- A similar precaution is needed in a system that uses the write-back protocol. Under this protocol, new data written into the cache are not written to the memory at the same time.
- Hence, data in the memory do not always reflect the changes that may have been made in the cached copy. It is important to ensure that such **stale data** in the memory are not transferred to the disk.
- One solution is to **flush the cache**, by forcing all dirty blocks to be written back to the memory before performing the transfer. The operating system can do this by issuing a command to the cache before initiating the DMA operation that transfers the data to the disk.
- Flushing the cache does not affect performance greatly, because such disk transfers do not occur often. The need to ensure that two different entities (the processor and the DMA subsystems in this case) **use identical copies of the data is referred to as a cache-coherence problem.**

Virtual Memory-1

99

- Recall that an important challenge in the design of a computer system is to provide a large, fast memory system at an affordable cost.
- Architectural solutions to increase the effective speed and size of the memory system.
- Cache memories were developed to increase the effective speed of the memory system.
- Virtual memory is an architectural solution to increase the effective size of the memory system.

Virtual Memory-2

100

- Recall that the addressable memory space depends on the number of address bits in a computer.
 - For example, if a computer issues 32-bit addresses, the addressable memory space is 4G bytes.
- Physical main memory in a computer is generally not as large as the entire possible addressable space.
 - Physical memory typically ranges from a few hundred megabytes to 1 G bytes.
- Large programs that cannot fit completely into the main memory have their parts stored on secondary storage devices such as magnetic disks.
 - Pieces of programs must be transferred to the main memory from secondary storage before they can be executed.

Virtual Memory-3

101

- When a new piece of a program is to be transferred to the main memory, and the main memory is full, then some other piece in the main memory must be replaced.
 - Recall this is very similar to what we studied in case of cache memories.

Virtual Memory-4

102

- Techniques that automatically move program and data between main memory and secondary storage when they are required for execution are called virtual-memory techniques.
- Processor issues binary addresses for instructions and data.
 - These binary addresses are called **logical or virtual addresses**.
- **Virtual addresses** are translated into **physical addresses** by a combination of hardware and software subsystems.
 - If virtual address refers to a part of the program that is currently in the main memory, it is accessed immediately.
 - If the address refers to a part of the program that is not currently in the main memory, it is first transferred to the main memory before it can be used.

Virtual Memory-5

103

- Memory management unit (MMU) **translates** virtual addresses into physical addresses.
- If the desired data or instructions are in the main memory they are fetched as described previously.
- If the desired data or instructions are not in the main memory, they must be transferred from secondary storage to the main memory.
- MMU causes the operating system to bring the data from the secondary storage into the main memory.

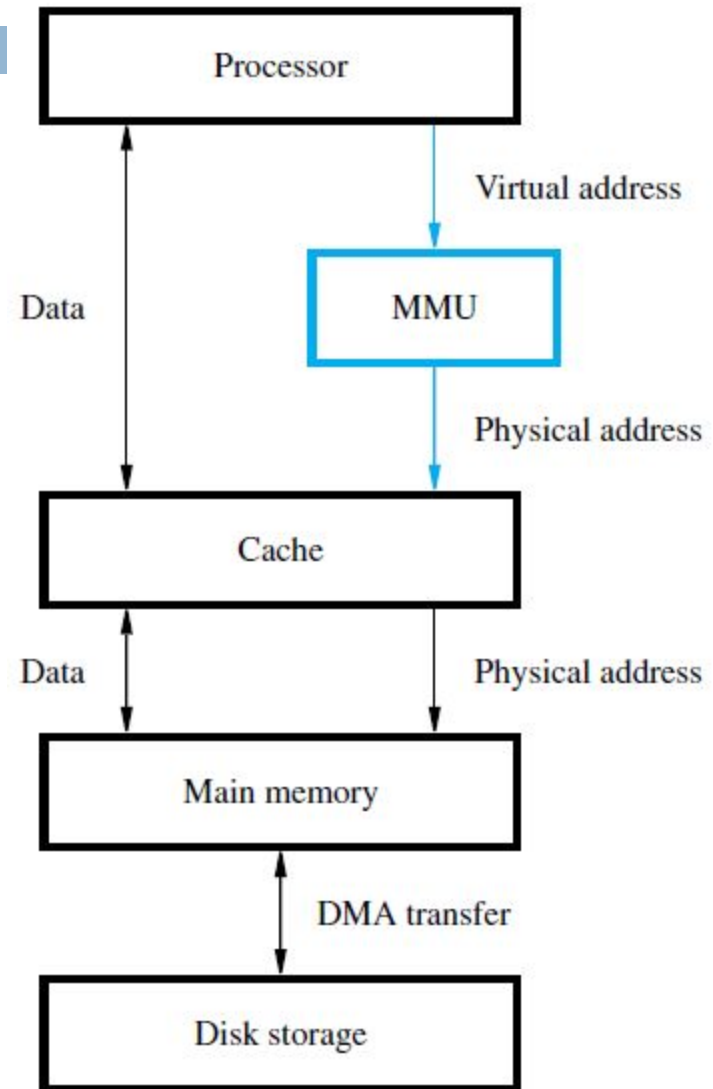


Figure 8.24 Virtual memory organization.

Virtual Memory-6

Address Translation

104

- Assume that program and data are composed of fixed-length units called pages.
- A page consists of a block of words that occupy contiguous locations in the main memory.
- Page is a basic unit of information that is transferred between secondary storage and main memory.
- Size of a page commonly ranges from 2K to 16K bytes.
 - Pages should not be too small, because the access time of a secondary storage device is much larger than the main memory.
 - Pages should not be too large, else a large portion of the page may not be used, and it will occupy valuable space in the main memory.

Virtual Memory-7

Address Translation

105

- Concepts of virtual memory are similar to the concepts of cache memory.

- Cache memory:
 - Introduced to bridge the speed gap between the processor and the main memory.
 - Implemented in hardware.

- Virtual memory:
 - Introduced to bridge the speed gap between the main memory and secondary storage.
 - Implemented in part by software.

Virtual Memory-8

Address Translation

106

- Each virtual or logical address generated by a processor is interpreted as a virtual page number (high-order bits) plus an offset (low-order bits) that specifies the location of a particular byte within that page.
- Information about the main memory location of each page is kept in the page table.
 - Main memory address where the page is stored.
 - Current status of the page.
- Area of the main memory that can hold a page is called as page frame.
- Starting address of the page table is kept in a page table base register.

Virtual Memory-9

Address Translation

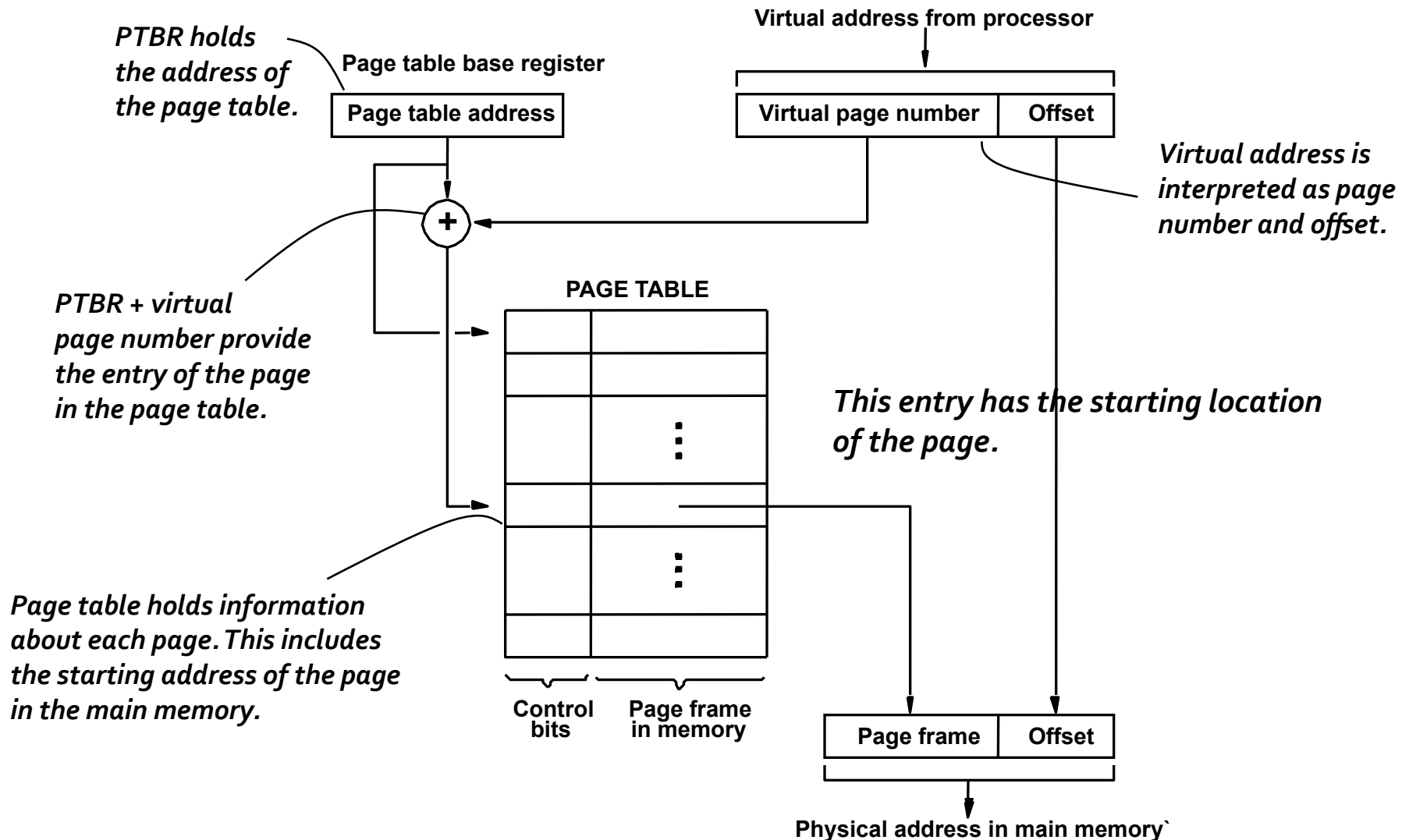
107

- Virtual page number generated by the processor is added to the contents of the page table base register.
 - This provides the address of the corresponding entry in the page table.
- The contents of this location in the page table give the starting address of the page if the page is currently in the main memory.

Virtual Memory-10

Address Translation

108



Virtual Memory-11

Address Translation

109

- Page table entry for a page also includes some control bits which describe the status of the page while it is in the main memory.
- One bit indicates the validity of the page.
 - Indicates whether the page is actually loaded into the main memory.
 - Allows the operating system to invalidate the page without actually removing it.
- One bit indicates whether the page has been modified during its residency in the main memory.
 - This bit determines whether the page should be written back to the disk when it is removed from the main memory.
 - Similar to the dirty or modified bit in case of cache memory.
- Other control bits for various other types of restrictions that may be imposed.
 - For example, a program may only have read permission for a page, but not write or modify permissions.

Virtual Memory-12

Address Translation

110

- Where should the page table be located?
- Recall that the page table is used by the MMU for every read and write access to the memory.
 - Ideal location for the page table is within the MMU.
- Page table is quite large.
- MMU is implemented as part of the processor chip.
- Impossible to include a complete page table on the chip.
- Page table is kept in the main memory.
- A copy of a small portion of the page table can be accommodated within the MMU.
 - Portion consists of page table entries that correspond to the most recently accessed pages.

Virtual Memory-13

Translation Lookaside Buffer

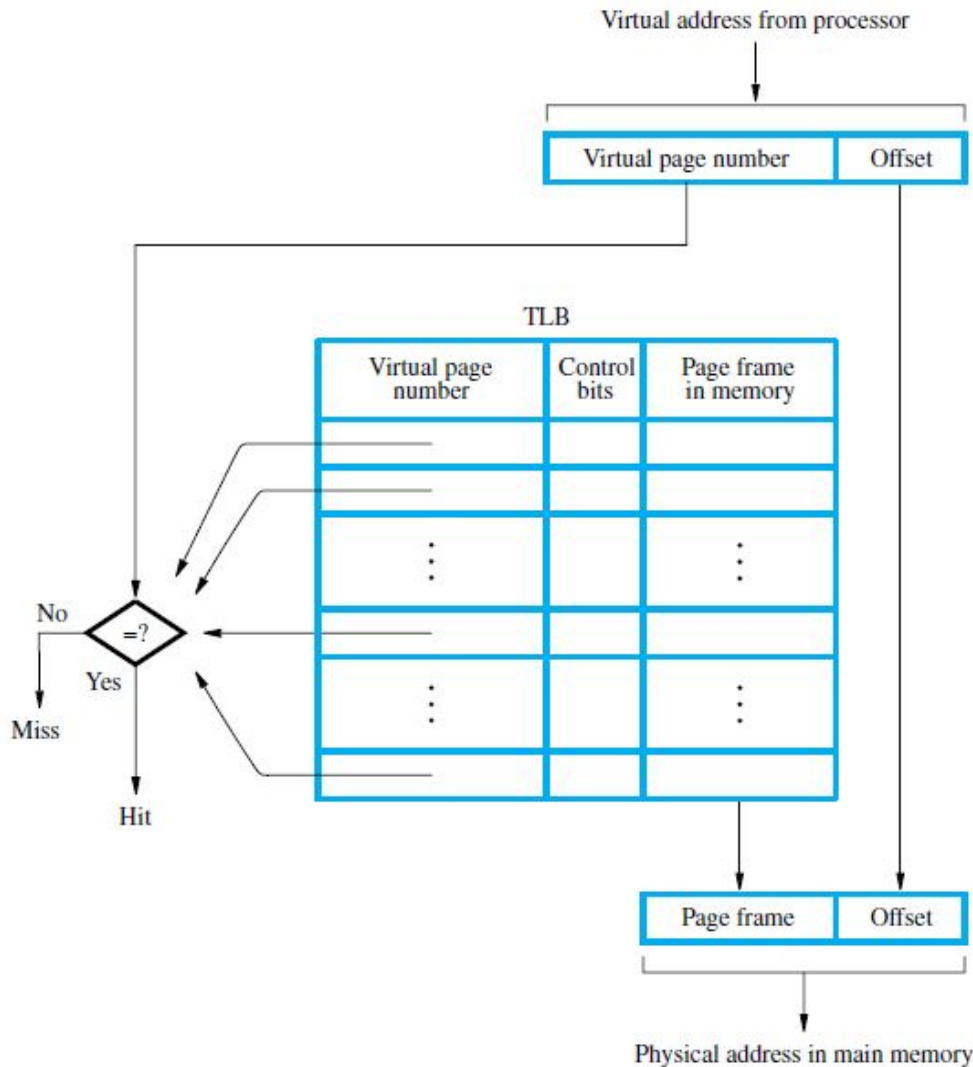
111

- A small cache called as Translation Lookaside Buffer (TLB) is included in the MMU.
 - TLB holds page table entries of the most recently accessed pages.
 - Operation of the TLB and page table in the main memory is similar to the operation of the cache and main memory.
- Page table entry for a page includes:
 - Address of the page frame where the page resides in the main memory.
 - Some control bits.
- In addition to the above for each page, TLB must hold the virtual page number for each page.

Virtual Memory-14

Translation Lookaside Buffer

112



- Associative-mapped TLB
- High-order bits of the virtual address generated by the processor select the virtual page.
- These bits are compared to the virtual page numbers in the TLB.
- If there is a match, a hit occurs and the corresponding address of the page frame is read.
- If there is no match, a miss occurs and the page table within the main memory must be consulted.
- Set-associative mapped TLBs are found in commercial processors.

Virtual Memory-15

Translation Lookaside Buffer

113

- How to keep the entries of the TLB coherent with the contents of the page table in the main memory?
- Operating system may change the contents of the page table in the main memory.
 - Simultaneously it must also invalidate the corresponding entries in the TLB.
- A control bit is provided in the TLB to invalidate an entry.
- If an entry is invalidated, then the TLB gets the information for that entry from the page table.
 - Follows the same process that it would follow if the entry is not found in the TLB or if a “miss” occurs.

Virtual Memory-16

Page Faults

114

- What happens if a program generates an access to a page that is not in the main memory?
- In this case, a page fault is said to occur.
 - Whole page must be brought into the main memory from the disk, before the execution can proceed.
- Upon detecting a page fault by the MMU, following actions occur:
 - MMU asks the operating system to intervene by raising an exception.
 - Processing of the active task which caused the page fault is interrupted.
 - Control is transferred to the operating system.
 - Operating system copies the requested page from secondary storage to the main memory.
 - Once the page is copied, control is returned to the task which was interrupted.

Virtual Memory-17

Page Faults

115

- Servicing of a page fault requires transferring the requested page from secondary storage to the main memory.
- This transfer may incur a long delay.
- While the page is being transferred the operating system may:
 - Suspend the execution of the task that caused the page fault.
 - Begin execution of another task whose pages are in the main memory.
- Enables efficient use of the processor.

Virtual Memory-18

Page Faults

116

- How to ensure that the interrupted task can continue correctly when it resumes execution?
- There are two possibilities:
 - Execution of the interrupted task must continue from the point where it was interrupted.
 - The instruction must be restarted.
- Which specific option is followed depends on the design of the processor.

Virtual Memory-19

Page Faults

117

- When a new page is to be brought into the main memory from secondary storage, the main memory may be full.
 - Some page from the main memory must be replaced with this new page.
- How to choose which page to replace?
 - This is similar to the replacement that occurs when the cache is full.
 - The principle of locality of reference can also be applied here.
 - A replacement strategy similar to LRU can be applied.
- Since the size of the main memory is relatively larger compared to cache, a relatively large amount of programs and data can be held in the main memory.
 - Minimizes the frequency of transfers between secondary storage and main memory.